Contextual Offline Demand Learning and Pricing with Separable Models

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This paper, inspired by a collaboration with a leading consumer electronics retailer in the Middle East, explores the challenge of demand learning and pricing using separable demand models. The data scarcity issue, characterized by limited price changes and low sales volumes, renders traditional models ineffective in deriving reasonable price elasticity. To address this issue, we advocate a separable model that leverages two submodels to distinctly capture the effects of price and contextual information. The separable structure enables us to invest special emphasis on the role of price and impose specific structural assumptions on the submodel for pricing effects, such as the monotone decreasing property. Theoretical analysis sheds light on the statistical complexity of demand learning with the separable structure, highlighting its capacity to reduce the necessary sample size to achieve a desired level of accuracy. We also introduce a computationally efficient iterative algorithm for deriving submodels from offline datasets, complete with convergence guarantees. In an empirical context, we demonstrate how our method can yield meaningful price elasticity estimations and revenue increase based on real sales data from the retailer.

Key words: Separable Model, Pricing, Contextual Information, Revenue Management

1. Introduction

In the past decade, leveraging the power of big data and analytics to enhance demand forecasting and pricing decisions has gained widespread recognition (see, e.g., Chase Jr 2013). However, despite the abundance of data, a great challenge is to transform the data into actionable insights for improved decision-making (Feng and Shanthikumar 2018a). In this work, we exemplify our collaboration with a leading retailer of consumer electronics in the Middle East to demonstrate how a retailer could extract reliable and useful demand information from a high-dimensional dataset,

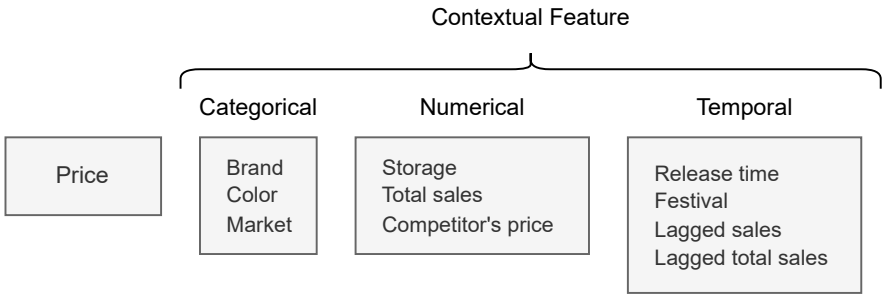


Figure 1 Input features of the MER's dataset

enabling future pricing optimization. To protect the confidentiality of this retail partner, we will refer to them throughout this paper as 'MER' (Middle East Retailer).

After completing the digitization process, MER has compiled purchasing historical data from each of its stores over the past three years on a cyclic basis, with each cycle typically spanning two days. Each piece of data is composed of 18 features, containing price, categorical features, numerical features and temporal features, presented in Figure 1. Our primary objective is to automate their pricing decisions for each cycle, drawing on insights gained from historical demand data. Yet, one of the most prominent characteristics of this historical data is its *scarcity*, both in terms of price changes and sales volumes. To provide context, we are specifically focusing on a distinct product category, mobile phones, for which we have collected data from 2000 SKUs (stock keeping units) spanning sales in the last three years. It's noteworthy that over 20% of these products maintained a single price throughout their entire life cycle, and more than half of all products experienced fewer than five price changes during their life cycle, as detailed in Table 1. This infrequency in price changes makes it exceedingly challenging to derive price elasticity information for each product individually. The situation is compounded by limited sales volume. Approximately 64% of these SKUs boast average per-cycle sales volumes of fewer than 1 unit, with approximately 83% having average per-cycle sales volumes of fewer than 5 units. The wide existence of low-sales products further restricts the amount of information that can be obtained from each SKU's data, particularly in terms of price elasticity. Consequently, learning price elasticity for each individual product proves nearly impossible due to the data's scarce nature. In addition, the scarcity issue does not seem to be unique for MER. Miao et al. (2022) observe a similar phenomenon for a giant Chinese online supermarket. To better illustrate the issue, we categorize all 2000 SKUs into four groups based on their price, referring to them as low, mid-low, mid-high, and high price groups.

As an initial attempt, within each price group, we apply a non-parametric regression-tree-based model on all input features, including price and contextual features. We conduct experiments using several standard techniques, such as the random forest regressor, LightGBM regressor (Ke et al.

Table 1 **Distribution of the number of different prices for one product**

# of different prices	Percentage
1	22%
2	8%
3	9%
4	12%
5+	49%

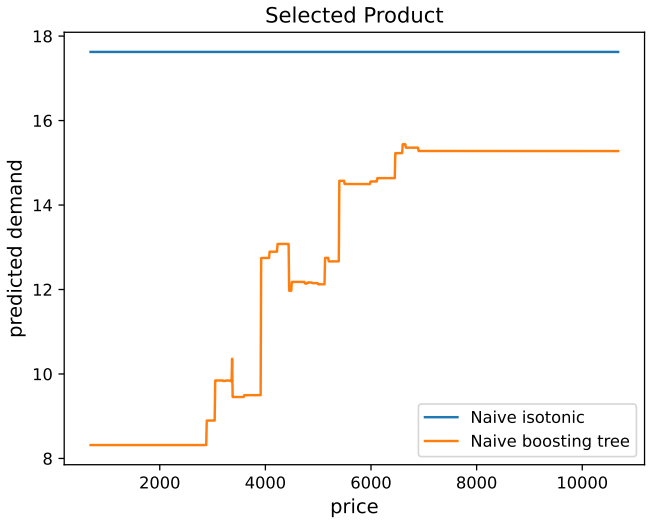


Figure 2 **Demand curve derived by two naive models.**

2017), and Catboost regressor (Prokhorenkova et al. 2018). While grouping the products into four categories helps mitigate the data scarcity issue to some extent, it is still insufficient because standard techniques yield demand curves below expectation. In fact, approximately 60% of the products exhibit a rising demand curve. For a representative product from the low-price group, as depicted in Figure 2, the traditional boosting tree algorithm predicts that quadrupling the price would somehow boost sales. This prediction contrasts with reality. Not only do these curves exhibit rising trends, but they also display unexplained and undesirable fluctuations.

One potential limitation of the non-parametric regression-tree-based model is its lack of domain knowledge concerning price elasticity. This model may overlook the distinct roles of pricing and the other contextual information, treating them as homogeneous features. This limitation has prompted us to incorporate business knowledge into the prediction model. In our collaboration with MER, we emphasize the role of price by *separating* it from other contextual features. This idea gives rise to the *separable model* framework. Specifically, we consider two variants: an additive separable model and a multiplicative separable model, as outlined below.

- Additive separable model:

$$\text{demand} = g_1(\text{price}) + g_2(\text{contextual feature}) + \text{error}.$$

- Multiplicative separable model:

$$\text{demand} = g_1(\text{price}) \times g_2(\text{contextual feature}) + \text{error}.$$

We are going to refer to g_1 and g_2 as *submodels*. In this framework, each segment of input feature is treated by its own model, allowing the flexibility to introduce domain knowledge and shape constraints. For example, we can constrain g_1 to be monotonically decreasing to obtain a decreasing demand curve. Slightly moving away from the MER example, another strong motivation to think about separable models is *data multimodality*. In many applications, the feature may include texts, images and sounds. It is reasonable to expect different kinds of features should require their own specialized submodels.

The central challenge is to utilize two distinct submodels to forecast demand using aggregated sales observations. In other words, we lack the direct observation of individual contributions from the submodels g_1 and g_2. A comparable challenge has been encountered in Nambiar et al. (2019) and Bu et al. (2022a), within the context of dynamic pricing. Nambiar et al. (2019) and Bu et al. (2022a) studied this problem in an online setting, where retailers can adaptively change pricing decisions, collecting new sales data on the fly. This flexibility offers opportunities to disentangle the effect of price and contextual information from aggregated sales observations, bypassing the challenge. Regrettably, we only have access to historical sales data, whose prices cannot be manipulated. Moreover, given that the true underlying demand model might lack a separable structure, we would like to consider the risk of *model mis-specification* as well.

1.1. Main Results and Contributions

In this paper, we introduce a novel demand learning model that employs a separable structure to integrate prior business knowledge into the demand model. Empirically, we demonstrate the practical application of this separable model in a real world demand learning problem. Theoretically, we comprehensively examine demand learning with a separable structure, both from a statistical and computational perspective. Our key theoretical findings can be summarized as follows:

- **Statistical analysis:** We derive the statistical sample complexity of the separable model under various assumptions, and compare it to models that do not employ the separable structure. In essence, we establish the number of samples required to learn a dependable separable structure through the traditional empirical risk minimizer. In cases where the actual demand model does exhibit a separable structure, we rigorously demonstrate the reduced sample requirements in our model compared to those without separable structures. Additionally, we show that the sample complexity we derive is applicable to a wide range of commonly-used function classes. The main technical challenge lies in understanding the covering number of the function class with a separable structure. Our analysis reveals that the logarithmic covering number of the separable function class can be upper-bounded by the sum of the logarithmic covering numbers of each of its submodels.

- **Computationally efficient algorithm:** In addition to achieving the desired statistical properties, we have developed a computationally efficient meta-algorithm known as *Submodel Descent* to attain the empirical risk minimizer. For each submodel, effective algorithms for empirical risk minimization are readily available, such as stochastic gradient descent for deep neural networks. However, given that we have only aggregated demand observations, it is not possible to optimize all submodels simultaneously. This algorithm employ a cyclic rule, updating each submodel sequentially while keeping all other submodels fixed, similar to a coordinate descent algorithm. Submodel Descent is compatible with a wide range of submodels and their corresponding optimization algorithms. We provide convergence guarantees of our Submodel Descent algorithm. With additional structural assumptions, we can even demonstrate (sub)linear convergence.

1.2. Related Work

Feature-based demand forecasts have been advocated in the long past, even before the era of big data. In the past, people from different industries tried to incorporate the domain knowledge in developing demand forecast, for example, the demographic information in retail (see, e.g., Hoch et al. 1995), and the weather index used in electricity consumption forecast (see, e.g., Carbonneau et al. 2008). More examples can be found in the study on semiconductor companies by Cakanyildirim and Roundy (2002) and in that on military distribution systems by Gardner (1990). More recently, a massive amount of data and information has become gradually accessible, especially with the rapid digitization of all industries. With the wave of digitization, massive data is becoming more and more accessible, which enables the success of advanced machine learning models in business analytics. Feng and Shanthikumar (2018b) introduce semi-nonparametric quantile regression method in learning supply and demand functions. Ferreira et al. (2016) apply the regression-tree based approach to conduct demand forecast and price optimization for an online retailer. Huber et al. (2019) adopt deep neural networks as a method for estimating demand. Lei et al. (2022a) introduce a transfer learning approach to predict the sales leveraging the power of data pooling. In general, model selection is the central question between data and intelligent decision. Simchi-Levi et al. (2023) propose a novel approach to calibrate the demand forecast during the pandemic. The separable structure considered in our paper can be a candidate when a decision-maker is choosing which model to use, and we hope that the model will be useful especially when the dataset is sparse and it is hard to get a good demand curve. Bu et al. (2022a) consider the price optimization with separable model is an online learning setting, where the decision maker has the flexibility to directly interact with the market to collect information adaptively and adjust the future price decisions. However, in our case, the data has already been collected and we do not have any control over the data collection process.

This paper is generally related to data-driven revenue management, see Chen et al. (2022a) and Chen and Hu (2023) for recent reviews. The most relevant line is pricing optimization with offline data. Bu et al. (2023) study the offline pricing problem with censored data under linear demand function without features. Qi et al. (2022) and Alley et al. (2023) consider the offline pricing problem with contextual features through a causal inference approach. Bu et al. (2022c) investigate the impact of pre-existing offline data on online dynamic pricing. Elmachtoub et al. (2023a) look into off-policy evaluation for a new personalized pricing policy based on historical data. Liu et al. (2023) propose a three-step semi-parametric estimation algorithm for offline feature-based pricing under a semi-parametric partially linear demand structure. Another line of work includes data-driven inventory problems or joint pricing and inventory control, see, e.g., Levi et al. (2007), Feng et al. (2014), Levi et al. (2015), Cheung and Simchi-Levi (2019), Qin et al. (2022), Lei et al. (2022b) and Ban and Rudin (2019). Additionally, assortment optimization is another topic in revenue management that is being reshaped by data; see, e.g., Talluri and Van Ryzin (2004), Kök et al. (2009), Rusmevichientong et al. (2010), Gallego and Topaloglu (2014), Gallego and Wang (2014), Miao and Chao (2020), Gao et al. (2021), Chen et al. (2022b). Moreover, dynamic pricing where the data is collected on the fly is gaining more and more attention as well; see, e.g., Broder and Rusmevichientong (2012), Keskin and Zeevi (2014), Qiang and Bayati (2016), Miao et al. (2019), Cohen et al. (2020), Chen and Gallego (2021), Wang et al. (2021), Keskin et al. (2022), Fan et al. (2022) and Li and Zheng (2023). In recent business practices, data-driven revenue management has been proved to be powerful and rewarding, see the case of Alibaba introduced by Deng et al. (2023).

In terms of methodology, our demand learning with separable models is closely related to ensemble learning, where multiple weak learning algorithms work collectively to obtain better predictive performance. For a general introduction, we refer to Zhang and Ma (2012). The fundamental difference is that each of our submodels is taking charge of a different segment of the features. However, ensemble learning rarely distinguishes the difference among the features.

2. Model Formulation

We consider a retailer who sells multiple products on its platform. The retailer keeps track of sales on a periodic basis, forming an offline dataset of n data points $\{(x_1, y_1), \cdots, (x_n, y_n)\}$. $x_i \in \mathcal{X} \subseteq \mathbb{R}^d$ records the features which comprehensively describe what the product is, where it is sold, and when does the sales occur. x_i can include price, categorical features (color, brand, etc.), numerical features (storage, market being sold, etc.) and temporal features (date, release time, etc.). The dimension d can be large in many scenarios, for example, in the case of the Middle East retailer, $d = 18$. $y_i \in \mathcal{Y} \subseteq \mathbb{R}$ is the corresponding sales volume, which follows a conditional probability measure $\mathbb{P}_{Y|X}$.

Let $\mathcal{F}$ be the model class of interest mapping $\mathcal{X}$ to $\mathcal{Y}$ and let $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ be a given loss function. The expected loss incurred by a model f is $\mathcal{L}(f) = \mathbb{E}\left[\ell(f(x), y)\right]$. Here, the expectation is taken with respect to all the randomness in the data generating process. The general objective is to find the "best" model $\tilde{f}$ that minimizes $\mathcal{L}(f)$ under the constraint that $\tilde{f} \in \mathcal{F}$. In this paper, we focus on the squared loss function $\ell(y', y) = (y' - y)^2$. Note that the best unconstrained model under square loss is the conditional expectation $f^*(x) = \mathbb{E}_{Y|X}\left[y \mid x\right]$ (Hastie et al. 2009). Under a finite dataset, a natural approach is to approximate the expected loss through empirical risk and seeking the empirical risk minimizer (ERM):

$$\hat{f}_n = \arg\min_{f \in \mathcal{F}} \hat{\mathcal{L}}(f) = \arg\min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \left(y_i - f(x_i)\right)^2. \tag{1}$$

Before prediction, the first fundamental task is model selection, essentially, to choose what kind of $\mathcal{F}$ to use. Model selection has been a central focus across various fields, such as computer science and operations research. Common selections for $\mathcal{F}$ include linear functions, boosting trees, and neural networks. Each model has its own set of advantages and drawbacks. Researchers, in general, try to strike a balance between the interpretability and predictive capabilities of these models based on their own tasks. In this paper, inspired by our collaborations with MER, we explore a new framework called *separable models*. In particular, we consider situations where input features can be separated into m disjoint segments, denoted as $x = (x^{(1)}, x^{(2)}, \cdots, x^{(m)})$. Such a segmentation proves to be quite common and natural in practical scenarios, especially when input features exhibit multimodality. For instance, these segments may correspond to textual, visual, and aural features. Consequently, it becomes reasonable to envision the treatment of each segment by a specialized model to capitalize on its inherent structure. Based on this intuition, we define separable models as $\mathcal{F} = \{h(g_1(x^{(1)}), \cdots, g_m(x^{(m)})) \mid g_j \in \mathcal{G}_j, j = 1, \cdots, m\}$, where $\mathcal{G}_j$ represents the specialized model class mapping the context segment $x^{(j)}$ to a real number, and $h : \mathbb{R}^m \to \mathbb{R}$ is a deterministic composition function mapping the recommendations from the m submodels to produce a final prediction. Thus, the ERM in Eq. (1) can be equivalently written as

$$\hat{f}_n = h(\hat{g}_1(x^{(1)}), \cdots, \hat{g}_m(x^{(m)})) = \arg\min_{\substack{g_j \in \mathcal{G}_j \\ j=1,\cdots,m}} \frac{1}{n} \sum_{i=1}^{n} (h(g_1(x_i^{(1)}), \cdots, g_m(x_i^{(m)})) - y_i)^2. \tag{2}$$

In this work, we emphasize the setting where feature segmentation, the choice of composition function, and the selection of submodel classes are predetermined. These choices are highly domain-specific and depend on the particular objectives and constraints of the prediction task. For instance, in the MER case, our goal extends beyond accurately forecasting future demand. We also aim to derive interpretable demand curves that can inform subsequent pricing optimizations. This additional objective motivates separating price as a distinct feature and applying an isotonic model with

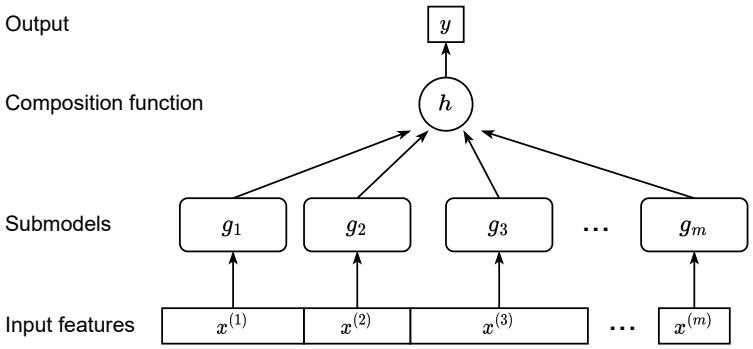


Figure 3 Architecture of the separable model.

shape constraints to capture its influence. Meanwhile, other features are treated with less specificity unless they become the focus of the analysis. For example, if the goal shifts to understanding the impact of categorical product attributes on demand, those features could be segmented and assigned their own submodel. Classical statistical techniques, such as cross-validation, could further aid in refining these segmentation and model selection decisions to best suit the task at hand.

EXAMPLE 1. In the context of MER, we separate the feature x_i into two distinct segments: (p_i, z_i). Here, p_i corresponds to the product's price, while z_i encapsulates various other contextual features. Each segment has its corresponding submodel. We employ isotonic regression to model the relationship between the price and sales volume. This method effectively allows us to capture the downward-sloping nature of demand curves typically associated with ordinary goods, i.e., $g_1(p_i) = \texttt{Iso}(p_i)$. On the other hand, we use boosting decision trees to model the intricate interplay between the remaining contextual features and the sales volume, i.e., $g_2(z_i) = \texttt{Tree}(z_i)$. To combine the suggestions from the two submodels, we consider two kinds of composite functions, the summation and the multiplication. More specifically,

- Additive separable model: $\mathcal{F}_{\text{add}} = \{\texttt{Iso}(p_i) + \texttt{Tree}(z_i)\}$. Notably, several well-established demand models can be considered as special cases of the additive separable model defined in this framework. For instance, the linear demand model $\mathcal{F}_{\text{linear}} = \{bp_i + c^\top z_i\}$ (see, e.g., Besbes and Zeevi 2015) and the semiparametric partial linear model $\mathcal{F}_{\text{semi-linear}} = \{bp_i + g(z_i)\}$ (see, e.g., Nambiar et al. 2019) fall into this category. Additive separable model should be used when we expect demand curves between products to exhibit parallel trends, meaning the demand curves have the same shape but differ by a constant.
- Multiplicative separable model: $\mathcal{F}_{\text{multip}} = \{\texttt{Iso}(p_i) \times \texttt{Tree}(z_i)\}$. Intuitively speaking, under this model $\texttt{Iso}(p_i)$ could be interpreted as the probability a customer purchases the product at price p_i, and $\texttt{Tree}(z_i)$ as the market size.

Practically, retailers can start with a richer composite functions and use the classical validation methods such as cross-validation to select the best one.

3. Statistical Analysis

The central research question in this section pertains to the statistical performance of $\hat{f}_n$ when given a dataset comprising n data points, especially in comparison to the optimal model, $\tilde{f}$. In other words, our objective is to investigate the convergence rate of $\hat{f}_n$ towards $\tilde{f}$ as the dataset size n increases.

Well-specified, fixed design. We first start with a *well-specified* setting i.e., $f^* \in \mathcal{F}$, and *fixed design*, where the input features $\{x_1, \cdots, x_n\}$ is fixed from the observed dataset. Note that although the input features are fixed, $\{y_1, \cdots, y_n\}$ are independently sampled based on the conditional probability measure $\mathbb{P}_{Y|X}$. We also define $\mathcal{F}|_{x_1,\cdots,x_n} = \{(f(x_1), f(x_2), \cdots, f(x_n)) \in \mathbb{R}^n : f \in \mathcal{F}\}$ as the restriction of the function class $\mathcal{F}$ to the fixed input features $x_1, \cdots, x_n$. Separating influence from noise, we can write the relation between x and y as:

$$y_i = f^*(x_i) + \eta_i, \tag{3}$$

where η_i is an independent zero-mean noise. We assume that all the noises η_i are σ^2-sub-Gaussian. Our goal is to study how well $\hat{f}_n$ can denoise the observed values and recover f^*. That is, our goal is to provide non-asymptotic bounds on the following random pseudometric for a given $\{x_1, \cdots, x_n\}$,

$$\mathbb{E}_\eta \|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P}_n)} := \mathbb{E}_\eta \left[\frac{1}{n} \sum_{i=1}^{n} (\hat{f}_n(x_i) - f^*(x_i))^2 \right], \tag{4}$$

where the expectation is over the random shocks $\eta_1, \cdots, \eta_n$ since $\hat{f}_n$ is randomized by them. Here, we define the L^2 norm of a function as $\|f\|_{L^2(\mathbb{P}_X)} = (\int_{\mathcal{X}} f(x)^2 \, d\mathbb{P}_X)^{(1/2)}$, so this metric coincides with the expectation of the L^2 norm under the empirical distribution $\mathbb{P}_n = (1/n) \sum_{i=1}^{n} \mathbb{1}_{\{x=x_i\}}$, thus having this abbreviation.

The complexity and richness of $\mathcal{F}$ is well known to have a direct impact on its statistical performance. To measure the complexity of a set with infinite cardinality, we introduce an essential concept called *covering number*.

Definition 1 (Covering number). (Wainwright (2019), Definition 5.1.) A *δ-cover* of a set $\mathbb{T}$ with respect to a metric ρ is a set $\{\theta_1, ..., \theta_K\} \subset \mathbb{T}$ such that for each $\theta \in \mathbb{T}$, there exists some $k \in 1, ..., K$ such that $\rho(\theta, \theta_k) \leq \delta$. The *$\delta$-covering number* $\mathcal{N}(\mathbb{T}, \rho, \delta)$ is the cardinality of the smallest δ-cover.

Roughly speaking, the δ-covering number is the smallest number of norm balls with radius δ that covers the entire set. In the same norm and for a given δ, a larger covering number signifies a "larger" and more intricate set. The covering number is a valuable measure for assessing the complexity of an infinite set, making it a useful tool for analyzing the complexity of the function class $\mathcal{F}$. The fundamental question explored in this work revolves around *how the separable structure influences the covering number of* $\mathcal{F}$. To delve into this query, we introduce a proposition that establishes a link between the complexity of the separable model class $\mathcal{F}$ and its submodel classes $\mathcal{G}_j$.

PROPOSITION 1. *Suppose that the composition function h is L-Lipschitz continuous under the Euclidean norm. Then we have*

$$\log \mathcal{N}(\mathcal{F}, L^2(\mathbb{P}_n), \delta) \leq \sum_{j=1}^{m} \log \mathcal{N}\left(\mathcal{G}_j, L^2(\mathbb{P}_n^{(j)}), \frac{\delta}{mL}\right), \tag{5}$$

where $\mathbb{P}_n^{(j)}$ is the empirical distribution of segment $x^{(j)}$, i.e., $\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}_{\{x^{(j)}=x_i^{(j)}\}}$.

Proposition 1 reveals how the separable structure contributes to reducing the structural complexity of $\mathcal{F}$. To demonstrate, for brevity, we adopt the notation $\texttt{Lip}(L,d)$ to denote the L-Lipschitz continuous function class on $[0,1]^d$. According to the standard results in textbooks (see, e.g., Wainwright 2019), $\log \mathcal{N}(\texttt{Lip}(L,d), L^2(\mathbb{P}_n), \delta) = \mathcal{O}((L/\delta)^d)$. Now, we compare it to a separable model class $\mathcal{F}_{\text{Lip}}$. Assuming that each of $\mathcal{F}_{\text{Lip}}$'s submodel class $\mathcal{G}_j$ is $\texttt{Lip}(L,d_j)$ and the composition function h is the additive function, it follows that $\mathcal{F}_{\text{Lip}}$ is a strict subset of $\texttt{Lip}(L,d)$. Without using Proposition 1, we obtain a loose upper bound on $\mathcal{F}_{\text{Lip}}$'s logarithmic covering number, $\mathcal{O}((L/\delta)^d)$, by looking at $\texttt{Lip}(L,d)$. Seemingly no improvement, however, if we use Proposition 1, because the addition function is 1-Lipschitz, we obtain a tighter upper bound $\mathcal{O}((mL/\delta)^{\max_j d_j})$. This means, compared to the non-separable model class $\texttt{Lip}(L,d)$, whose logarithmic covering number is of order $\mathcal{O}((L/\delta)^d)$, the separable structure reduces the covering number to at most $\mathcal{O}((mL/\delta)^{\max_j d_j})$. This outcome underscores that the logarithmic covering number of the separable model is predominantly determined by the maximum dimension among d_j, which may be substantially smaller than the total dimension d. Another commonly-used function class in both academia and industry is linear function class, i.e., $\texttt{Lin}(d) = \{x \mapsto \langle w, x\rangle \mid x \in \mathbb{R}^d, \|w\| \leq 1\}$, whose covering number can be controlled by $\log \mathcal{N}(\texttt{Lin}(d), L^2(P_n), \delta) = \mathcal{O}(d\log(1/\delta))$. If $\mathcal{G}_j$ is $\texttt{Lin}(d_j)$, from Proposition 1 we can derive $\log \mathcal{N}(\mathcal{F}, L^2(\mathbb{P}_n), \delta) = \sum_{j=1}^{m} \mathcal{O}(d_j \log(1/\delta)) = \mathcal{O}(d\log(1/\delta))$, which is exactly of the same order of $\texttt{Lin}(d)$. This coincides with the fact that the overall model class $\mathcal{F}$ is exactly the linear model class $\texttt{Lin}(d)$. This observation aligns with the intuition that the linear function class inherently possesses a separable structure, hence no improvements will be made by introducing the separable structure. The example of linear model also shows that the bound given by Proposition 1 is tight ignoring lower order terms and constants.

REMARK 1. In general, separating nonparametric models results in a significant reduction in complexity, whereas separating parametric classes may not yield the same level of complexity reduction. Rakhlin (2022) formally defines the parametric and non-parametric model class by the structure of their covering number. According to this definition, a model class $\mathcal{F}$ is defined to be parametric if $\log \mathcal{N}(\mathcal{F}, L^2(\mathbb{P}_n), \delta) = \mathcal{O}(d\log(1/\delta))$ for some dimension parameter d, and nonparametric if $\log \mathcal{N}(\mathcal{F}, L^2(\mathbb{P}_n), \delta) = \mathcal{O}((1/\delta)^\alpha)$ for some $\alpha > 0$. Notably, the value of α increases with the

dimension of the domain, indicating an exponential dependence for nonparametric classes, in contrast to the linear dependence seen in parametric classes. When $\mathcal{F}$ is parametric, separation does not influence complexity because $\mathcal{O}(d\log(1/\delta)) = \mathcal{O}(\sum_{j=1}^{m} d_j \log(1/\delta))$. When $\mathcal{F}$ is nonparametric, separation would reduce complexity from $\mathcal{O}(\log(1/\delta)^{\alpha})$ to $\mathcal{O}(\log(1/\delta)^{\max_j \alpha_j})$. This agrees with the prior discussion regarding L-Lipschitz and linear function classes.

Based on the definition of covering number and Proposition 1, we present the main theorem, which gives an upper bound on the statistical complexity of the offline learning problem. This theorem is derived based on Liang et al. (2015).

THEOREM 1. *Assume $\{x_1, \cdots, x_n\}$ is fixed, $\{\eta_1, \cdots, \eta_n\}$ is independent and σ^2-subGaussian and $y_i = f^*(x_i) + \eta_i$ with $f^* \in \mathcal{F}$, we have*

$$\begin{aligned}&\mathbb{E}_\eta \|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P}_n)} \\ &\quad \le \inf_{\delta \ge 0} 4\sigma \left\{ \frac{\sum_{j=1}^{m} \log \mathcal{N}(\mathcal{G}_j, L^2(\mathbb{P}_n), \frac{\delta}{mL})}{2\sigma n} + \frac{12}{\sqrt{n}} \int_0^{\delta} \sqrt{\sum_{j=1}^{m} \log \mathcal{N}\left(\mathcal{G}_j, L^2(\mathbb{P}_n), \frac{\epsilon}{mL}\right)} d\epsilon \right\}, \end{aligned} \tag{6}$$

if the composition function h is L-Lipschitz continuous under the Euclidean norm.

Both terms in the RHS of Eq. (6) are intricately linked to the parameter δ. The first term characterizes the statistical complexity of identifying the model within the δ-cover that best approximates f^*. Selecting a smaller δ results in a larger covering set. Consequently, when δ becomes smaller, this identification task is more difficult, leading to a larger first term. The second term quantifies the approximation error incurred when representing $\mathcal{F}$ with a δ-cover. When δ is smaller, the δ-cover is finer, and the approximation error is smaller. Therefore we have two terms that goes against each other, and the minimum upper bound, as provided by Equation (6), represents the ideal selection of δ that trades off between the identification complexity and approximation error.

Under the criterion of minimax optimality in Yang and Barron (1999), it can be shown that the upper bound is generally tight except for a corner case when the separable model class is too complex. More specifically, the statistical rate gap can possibly occur when $\log \mathcal{N}(\mathcal{F}, L^2(\mathbb{P}_n), \delta) = \Omega(\delta^{-\alpha})$ and $\alpha \ge 2$. Many commonly used function classes, for example, the Lipschitz function class $\texttt{Lip}(L, d)$ and the linear function class $\texttt{Lin}(d)$, fall into the well-behaved region (i.e. $\alpha < 2$). Additionally, the mismatch is not a flaw of the composite structure, but from a deeper root that is the suboptimality of ERM under square loss and highly complex model classes, see Kur et al. (2020) for more details.

To illustrate the blessing of the separable structure, we turn our attention back to the Lipschitz function class $\texttt{Lip}(L, d)$ with additive separation. Note that Theorem 1 remains applicable to the

general $\texttt{Lip}(L,d)$ without any additional separable structure, simply by choosing $m=1$. With its δ-covering number being $\mathcal{O}((L/\delta)^d)$, we take $\delta=\mathcal{O}(n^{-\frac{1}{d+2}})$, which balances the two terms in Theorem 1 and yields a rate-optimal upper bound, i.e., $\mathbb{E}\|\hat{f}_n-f^*\|^2_{L^2(\mathbb{P}_n)}=\mathcal{O}(n^{-\frac{2}{d+2}})$. Simultaneously, by following a similar analytical approach, the additive separable model class attains a statistical performance of $\mathcal{O}(n^{-\frac{2}{\max_j d_j+2}})$. In an extreme case where $d_1=d_2=\cdots=d_m=d/m$, the separable structure results in $\mathcal{O}(n^{-\frac{2}{d/m+2}})$, much smaller than $\mathcal{O}(n^{-\frac{2}{d+2}})$ when m is large. In other words, one immediate advantage of utilizing separable models is the reduction in sample complexity. Specifically, in order to achieve $\mathbb{E}\|\hat{f}_n-f^*\|^2_{L^2(\mathbb{P}_n)}\le\varepsilon$, our separable model is only asking for roughly $\varepsilon^{-\frac{d/m+2}{2}}$ samples, while the model without considering the separable structure demands as many as $\varepsilon^{-\frac{d+2}{2}}$ samples. We note that in Remark 1, the separable structure may not significantly reduce the covering number needed for parametric model classes, and thus may not help with the reduction of sample complexity. We conclude with Table 2, in which we compare the L-Lipschitz model with its separable variants, further illustrating the statistical advantages derived from the separable structure.

Table 2 Comparison on the Statistical Performance of Separable Models

Model	Assumption	Performance
$f(x)$	$f\in\texttt{Lip}(L,d)$	$\mathcal{O}(n^{-\frac{2}{d+2}})$
$f=g_1(x^{(1)})+g_2(x^{(2)})$	$g_j\in\texttt{Lip}(L,d_j)$ for $j=1,2$	$\mathcal{O}(n^{-\frac{2}{(d_1\vee d_2)+2}})$
$f=\sum_{j=1}^m g_j(x^{(j)})$	$g_j\in\texttt{Lip}(L,d_j)$ for $j=1,\cdots,m$	$\mathcal{O}(n^{-\frac{2}{\max(d_1,\cdots,d_m)+2}})$
$f=\prod_{j=1}^m g_j(x^{(j)})$	Same as above	Same as above

Mis-specified, fixed design. Model specification is usually a strong assumption. From a pessimistic perspective, we should always be prepared for a *mis-specified* setting when $\mathcal{F}$ fails to capture f^* ($f^*\notin\mathcal{F}$, or equivalently, $\tilde{f}\neq f^*$). Now we move on to the mis-specified case.

Since f^* is no longer possible to obtain in the mis-specified setting, a reasonable target that we should aim at is $\tilde{f}\in\arg\min_{f\in\mathcal{F}}\|f-f^*\|^2_{L^2(\mathbb{P}_n)}$. $\tilde{f}$ is one of the best estimators one can acquire under the constraint $f\in\mathcal{F}$. This leads to the metric of our interest,

$$\mathbb{E}_\eta\|\hat{f}_n-f^*\|_{L^2(\mathbb{P}_n)}-\|\tilde{f}-f^*\|^2_{L^2(\mathbb{P}_n)}, \tag{7}$$

which is also referred to as *excess risk* in the literature. In this metric, we measure the gap between the performance of the ERM's, and the best estimator's performance. The following theorem describes the excess risk upper bound of the ERM.

THEOREM 2. *For the fixed design and mis-specified case, suppose that the composition function is L-Lipschitz and $\mathcal{F}|_{x_1,\cdots,x_n}$ is convex, we have*

$$\mathbb{E}_\eta\|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P}_n)} - \|\tilde{f} - f^*\|^2_{L^2(\mathbb{P}_n)} \leq \inf_{\delta\geq 0} 2\sigma\left\{\frac{\sum_{j=1}^m \log\mathcal{N}(\mathcal{G}_j, L^2(\mathbb{P}_n), \frac{\delta}{mL})}{\sigma n} + \frac{12}{\sqrt{n}}\int_0^\delta \sqrt{\sum_{j=1}^m \log\mathcal{N}\left(\mathcal{G}_j, L^2(\mathbb{P}_n), \frac{\epsilon}{mL}\right)}d\epsilon\right\}.$$

We would like to highlight the role of the convexity assumption. Comparing Theorems 1 and 2, we observe that, under the convexity assumption, the convergence rate in the mis-specified setting closely mirrors that of the well-specified scenario. This implies that the task is not more difficult for a convex model class. The convexity assumption for $\mathcal{F}$ encompasses a wide range of commonly-used assumptions, such as the linear function class, the convex function class, the bounded function class, and the Lipschitz function class. Consequently, the results presented in Table 2 can be seamlessly extended to the mis-specified setting, given that the considered $\mathcal{F}$ in the table is a convex set. When $\mathcal{F}$ is not convex, it turns out the ERM estimator itself could be suboptimal, even when $\mathcal{F}$ is finite. A patch of this proposes the star estimator, which slightly modifies the ERM estimator to make it optimal again. We refer to Audibert (2007) for further details.

Mis-specified, random design. We conclude this section with the most general setting, the mis-specified and random design case. By random design, we no longer see $\{x_1,\cdots,x_n\}$ as fixed but randomly generated from a probability measure $\mathbb{P}_X$. The metric becomes

$$\mathbb{E}_{X,\eta}\|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P}_X)} - \|\tilde{f} - f^*\|^2_{L^2(\mathbb{P}_X)},$$

where the "best" constrained model in this case is defined as $\tilde{f} \in \arg\min_{f\in\mathcal{F}}\|f - f^*\|^2_{L^2(\mathbb{P}_X)}$. Compared to the metric of fixed design in Eq. (7), the first difference is that the L^2 norm is no longer taken under the empirical distribution, but under the actual distribution $\mathbb{P}_X$. The second difference is now $\hat{f}_n$ is randomized by not only the shocks, but also the features. Therefore the expectation is taken with respect to both η_i and x_i.

Preceding the theorem, we introduce the uniform boundedness assumption. We assume that there exists a constant $C>0$ such that $|f|\leq C$ for all $f\in\mathcal{F}$. Without loss of generality, we can let $C=1$.

THEOREM 3. *For the mis-specified, random design case, suppose that the composition function is L-Lipschitz and $\mathcal{F}$ is convex and uniformly bounded by 1, we have, with probability at least $1-\zeta$,*

$$\mathbb{E}_{X,\eta}\|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P})} - \|\tilde{f} - f^*\|^2_{L^2(\mathbb{P})} \leq \left(\mathbb{E}_\eta\|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P}_n)} - \inf_{f\in\mathcal{F}}\|f - f^*\|^2_{L^2(\mathbb{P}_n)}\right) + \\ + \frac{48}{\sqrt{n}}\int_0^1 \sqrt{\sum_{j=1}^m \log\mathcal{N}\left(\mathcal{G}_j, L^2(\mathbb{P}_n), \frac{\epsilon}{mL}\right)}d\epsilon + 4\sqrt{\frac{2\log(2/\zeta)}{n}}. \quad (8)$$

Equation (8) consists of three terms in the RHS. The first term corresponds to the loss metric for the mis-specified, fixed design case. Because the features $\{x_1, \cdots, x_n\}$ are random, the empirical measure $\mathbb{P}_n$ is random, and the loss metric is essentially a random variable determined by $\{x_1, \cdots, x_n\}$. Nevertheless, Theorem 2 establishes a uniform upper bound regardless of the realization of $\{x_1, \cdots, x_n\}$, therefore we can replace this term with that upper bound. The second term represents the incurred error by approximating the random design metric with the fixed design one, and the scale of this approximation error depends on the complexity of $\mathcal{F}$. The last term arises due to the random sampling of x_is. It is worth noting that, regardless of the structure of $\mathcal{F}$, both the last two terms in Eq.(8) contains a $1/\sqrt{n}$ term, invariably exhibiting an order of $\mathcal{O}(1/\sqrt{n})$. Consequently, the convergence rate for the random design parallels that of the fixed design, provided that the rate is slower than $\mathcal{O}(1/\sqrt{n})$. When the fixed design achieves a faster rate, the randomness of features would kick in, bottlenecking the rate to $\mathcal{O}(1/\sqrt{n})$.

We provide some additional comments on the uniform boundedness assumption. This assumption is introduced to ensure that the empirical norm $\|f\|_{L^2(\mathbb{P}_n)}$ for all $f \in \mathcal{F}$ remains relatively close to the population norm $\|f\|_{L^2(\mathbb{P}_X)}$. It is possible to relax this assumption. Such relaxation necessitates a more intricate analysis, and it has been thoroughly explored in Mendelson (2015).

Finally, Theorem 3 provides important insights into the famous *bias-variance trade-off* problem. When the true data generating process is not necessarily separable, the mis-specified, random design setting presents a fundamental tension between model expressiveness and statistical efficiency. In this context, bias emerges from potential model mis-specification, quantified by the performance gap between the best constrained model and the true model: $\|\tilde{f} - f^*\|^2_{L^2(\mathbb{P}_X)}$. Variance manifests in the statistical performance of the empirical risk minimizer $\hat{f}_n$, measured by the sample complexity term $\mathbb{E}_{X,\eta}\|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P})} - \|\tilde{f} - f^*\|^2_{L^2(\mathbb{P})}$.

This trade-off becomes clear when we consider expanding the model class $\mathcal{F}$. A larger model class reduces the risk of mis-specification by allowing $\tilde{f}$ to better approximate the true model f^*. However, this comes at the cost of increased sample complexity, as evidenced by larger covering numbers in our theoretical bounds, resulting in slower convergence of the empirical risk minimizer. Conversely, restricting the model class improves statistical efficiency but potentially introduces larger approximation errors through increased model mis-specification.

The consideration of bias-variance trade-off is one of the deeper motivations of the separable model framework, particularly in contexts like the demand learning and price optimization problem with MER. The failure of classical methods, as illustrated in Figure 2, suggests that the available data may be insufficient to reliably train highly complex models. To obtain meaningful demand curves, we must introduce additional structure that effectively reduces model class complexity. Our simulation studies with synthetic data further support this conclusion. In the study, we compare

three models of increasing complexity: linear models, feature-segmented additive separable models, and fully connected neural networks. The simulation results demonstrate that separable models achieve superior generalization performance on test data when training data is limited. Detailed results of these simulations are provided in the appendix.

4. Optimization and Computational Analysis

In the previous section, we have established the statistical property of the ERM $\hat{f}_n$. A remaining question is how to numerically compute $\hat{f}_n$ given a dataset $\{(x_1, y_1), \cdots, (x_n, y_n)\}$ in hand. In other words, we want to find an efficient algorithm to solve the optimization problem $\arg\min_{f\in\mathcal{F}} \hat{\mathcal{L}}(f) = \arg\min_{f\in\mathcal{F}} (1/n)\sum_{i=1}^n (f(x_i) - y_i)^2$. The opportunity, as well as challenge, brought by our separable structure is that for each individual submodel $\mathcal{G}_j$, there may already exist handy optimization algorithms, such as ordinary least squares for linear models and stochastic gradient descent for neural networks. However, we only have an aggregated observation of y_i, but do not know exactly how much each submodel contributes to y_i. What is even worse, the estimation error of each submodel will couple with each other. To further motivate our argument, let us consider the MER's case. As shown in Example 1, we use two well-studied model classes as submodels: isotonic regression (`Iso`) and boosting decision trees (`Tree`). For `Iso`, a long history exists dating all the way back to 1950s (see, e.g., van Eeden 1958). We refer to Groeneboom and Jongbloed (2014) for a general introduction. One simple algorithm is called the *PAVA*, or *Pool Adjacent Violators Algorithm* (Brunk et al. 1972, Chakravarti 1989). For `Tree`, since the seminal work on gradient boosting in Friedman (2001), there have been many publicly available packages on the gradient boosting tree model, including XGBoost (Chen and Guestrin 2016), LightGBM (Ke et al. 2017), and Catboost (Prokhorenkova et al. 2018). Thus, both `Iso` and `Tree` model classes themselves have efficient and easy-to-use implementations. However, what we are interested in is the composition of the two models, rather than using them individually. This requires us to integrate the efficient algorithms together, and train the two models collectively.

To this end, we propose and analyze a simple algorithm called *Submodel Descent*, taking inspirations from the well-known coordinate descent algorithm. We first formally define the optimization problem. Our focus in this section is to devise a routine that efficiently solves the following optimization task:

$$\min_{g_j\in\mathcal{G}_j, \forall j=1,\cdots,m.} \quad \hat{\mathcal{L}}(f) = \hat{\mathcal{L}}(g_1, \cdots, g_m) = \frac{1}{n}\sum_{i=1}^n (h(g_1(x_i^{(1)}), \cdots, g_m(x_i^{(m)})) - y_i)^2. \tag{9}$$

Here we abuse the notation for brevity and write $\hat{\mathcal{L}}(f)$ and $\hat{\mathcal{L}}(g_1, \cdots, g_m)$ interchangeably, since the loss incurred by composite model f is determined by its submodels g_j. We assume the existence of algorithmic *oracles*, which are subroutines that can find the exact or approximate optimal solution

for the submodels. We will discuss two scenarios, *exact oracles* and *inexact oracles*. Exact oracles find exact solutions of the their corresponding submodels. Examples would include ordinary least squares and the PAVA algorithm, which yield solutions up to machine precision. Nevertheless, in practice, a lot of subroutines are iterative algorithms, which only finds an approximate solution under finite running time. Therefore we also consider inexact oracles that only provide approximate solutions. Stochastic gradient descent and algorithms for gradient boosting trees would fall within this category.

ASSUMPTION 1 **(Exact Oracle)**. *Each submodel class $\mathcal{G}_j$ is associated with an exact algorithmic orcale, which is a subroutine that exactly finds the optimal g_j of the following problem:*

$$g_j \in \arg\min_{\gamma_j \in \mathcal{G}_j} \hat{\mathcal{L}}(g_1, \cdots, g_{j-1}, \gamma_j, g_{j+1}, \cdots, g_m), \tag{10}$$

for any given $g_k \in \mathcal{G}_k$, $k = 1, \cdots, j-1, j+1, \cdots, m$.

ASSUMPTION 2 **(Inexact Oracle)**. *Each submodel class $\mathcal{G}_j$ is associated with an inexact algorithmic orcale, which is a subroutine that finds an approximate solution $\tilde{g}_j$ of Eq.(10), ensuring that this solution is no more than ε-distance from the nearest optimal solution:*

$$\|\tilde{g}_j - \arg\min_{\gamma_j \in \mathcal{G}_j} \hat{\mathcal{L}}(g_1, \cdots, g_{j-1}, \gamma_j, g_{j+1}, \cdots, g_m)\|_{L^2(\mathbb{P}_n)} \leq \varepsilon \tag{11}$$

for any given $g_k \in \mathcal{G}_k$, $k = 1, \cdots, j-1, j+1, \cdots, m$. Additionally, we require that this inexact oracle must reduce the objective loss function with each application.

Assumptions 1 and 2 effectively treats each submodel as a black box, with all training on the submodel being conducted solely through its oracle. This approach simplifies our implementation and analysis, as it can be easily generalized to different submodels and subroutines. Additionally, this perspective allows us to regard each submodel as a *coordinate*, enabling us to optimize each coordinate individually. From this standpoint, we introduce a straightforward yet efficient algorithm in Algorithm 1 that is compatible with different types of submodels. This algorithm integrates the oracles of each submodel and facilitates the sequential training of all submodels. In the sequel, we first focus on the ideal case of exact oracles to understand the performance of our algorithm, then we analyze the stability of the algorithm under inexact oracles with an error margin.

Exact Oracle. In Algorithm 1, each iteration follows the cyclic rule, sequentially updating the submodels. Treating each submodel as a coordinate, Algorithm 1 can be viewed as a cyclic coordinate descent algorithm applied to a functional $\hat{\mathcal{L}}(f)$. To delve deeper into this perspective, it is essential to recognize that, for each g_j, only the values $(g_j(x_1), \cdots, g_j(x_n))$ at the data points $x_1, \cdots, x_n$ have any impact on the loss, rendering the model values at other points irrelevant.

Algorithm 1: Submodel Descent

Input. $\{(x_i, y_i)\}_{i=1}^n$; $\{\mathcal{G}_j\}_{j=1}^m$; h; ℓ; T.

Initialization. Initialize with $(g_1^0, \cdots, g_m^0)$.

for *Iteration* $t = 1, ..., T$ **do**

Cyclically update the submodels with their respective oracles:

$$\hat{g}_1^t \in \arg\min_{\gamma_1 \in \mathcal{G}_1} \hat{\mathcal{L}}(\gamma_1, \hat{g}_2^{t-1}, \cdots, \hat{g}_m^{t-1}),$$
$$\hat{g}_2^t \in \arg\min_{\gamma_2 \in \mathcal{G}_2} \hat{\mathcal{L}}(\hat{g}_1^t, \gamma_2, \hat{g}_3^{t-1}, \cdots, \hat{g}_m^{t-1}),$$
$$\cdots$$
$$\hat{g}_m^t \in \arg\min_{\gamma_m \in \mathcal{G}_m} \hat{\mathcal{L}}(\hat{g}_1^t, \hat{g}_2^t, \cdots, \hat{g}_{m-1}^t, \gamma_m).$$

Output. $\hat{f}^T = h(\hat{g}_1^T, \cdots, \hat{g}_m^T)$.

Consequently, it becomes sufficient to consider the restriction of $\mathcal{G}$ to the input points $x_1, \cdots, x_n$, denoted as $\mathcal{G}_j|_{x_1,\cdots,x_n} \subseteq \mathbb{R}^n$, rather than the entire model class $\mathcal{G}_j$. For the sake of clarity, we denote g_j's value at x_i as $g_{j,i}$, the vector of values across all inputs $(g_{j,1}, \cdots, g_{j,n})$ as $\mathbf{g}_j$, and the $\mathbb{R}^{m\times n}$ vector formed by concatenating $(\mathbf{g}_1, \cdots, \mathbf{g}_m)$ as $\mathbf{g}$. The functional optimization problem described in Eq. (9) is equivalent to the following optimization problem defined within the vector space $\mathbb{R}^{m\times n}$.

$$\begin{aligned} \text{minimize} \quad & \hat{\mathcal{L}}(\mathbf{g}_1, \cdots, \mathbf{g}_m) = \frac{1}{n}\sum_{i=1}^n (h(g_{1,i}, \cdots, g_{m,i}) - y_i)^2 \\ \text{subject to} \quad & \mathbf{g}_j \in \mathcal{G}_j|_{x_1,\cdots,x_n}, \quad \forall j = 1, \cdots, m. \end{aligned} \tag{12}$$

Here again, we abuse the notation and write $\hat{\mathcal{L}}(\mathbf{g}_1, \cdots, \mathbf{g}_m)$ to underscore that only the submodels' values at dataset inputs influence the loss. Applying the same procedure as outlined earlier, Algorithm 1 can be transformed into an equivalent cyclic block coordinate descent algorithm, presented in Algorithm 2. Algorithm 2 serves as an important theoretical instrument to analyze Algorithm 1's computational performance. It is worth noting that we concentrate on the cyclic update rule due to its simplicity. Alternative update rules are also feasible. These may include random sampling (where the coordinates to be updated are sampled with replacement) and random permutation (where the coordinates are sampled without replacement).

EXAMPLE 1 (CONTINUED). We demonstrate the specific implementation of Algorithm 1 in the MER's case in Example 1. In practice, Submodel Descent effectively performs repeated regressions of the current residual onto the submodel being updated, helping to partition the overall variation in demand.

- Additive separable model: $\mathcal{F}_{\text{add}} = \{\texttt{Iso}(p_i) + \texttt{Tree}(z_i)\}$. In each iteration t, we first calculate the residual of $\texttt{Tree}^{t-1}$ as $r_{i,\texttt{Tree}}^t = y_i - \texttt{Tree}^{t-1}(x_i)$. Then, we use the PAVA (`Iso`'s oracle) to

Algorithm 2: Coordinate Descent (Algorithm 1's equivalent form)

Input. $\{(x_i, y_i)\}_{i=1}^n$; $\{\mathcal{G}_j\}_{j=1}^m$; h; l; T.

Initialization. Initialize with $\hat{\mathbf{g}}^0 = (\hat{\mathbf{g}}_1^0, \cdots, \hat{\mathbf{g}}_m^0)$.

for *Iteration* $t = 1, ..., T$ **do**

Cyclically update the blocks:

$$\hat{\mathbf{g}}_1^t \in \underset{\gamma_1 \in \mathcal{G}_1|_{x_1,\cdots,x_n}}{\arg\min} \hat{\mathcal{L}}(\gamma_1, \hat{\mathbf{g}}_2^{t-1}, \cdots, \hat{\mathbf{g}}_m^{t-1}),$$
$$\hat{\mathbf{g}}_2^t \in \underset{\gamma_2 \in \mathcal{G}_2|_{x_1,\cdots,x_n}}{\arg\min} \hat{\mathcal{L}}(\hat{\mathbf{g}}_1^t, \gamma_2, \hat{\mathbf{g}}_3^{t-1}, \cdots, \hat{\mathbf{g}}_m^{t-1}),$$
$$\cdots$$
$$\hat{\mathbf{g}}_m^t \in \underset{\gamma_m \in \mathcal{G}_m|_{x_1,\cdots,x_n}}{\arg\min} \hat{\mathcal{L}}(\hat{\mathbf{g}}_1^t, \hat{\mathbf{g}}_2^t, \cdots, \hat{\mathbf{g}}_{m-1}^t, \gamma_m).$$

Output. $\hat{\mathbf{g}}^T = (\hat{\mathbf{g}}_1^T, \cdots, \hat{\mathbf{g}}_m^T)$.

fit the isotonic model to the dataset $\{(x_i, r_{i,\texttt{Tree}}^t)\}_{i=1}^n$, acquiring $\texttt{Iso}^t$. Fitting the residual dataset under square loss is equivalent to solving Eq. (10) with the `Tree` model being fixed. After that, we calculate the residual of $\texttt{Iso}^t$ as $r_{i,\texttt{Iso}}^t = y_i - \texttt{Iso}^t(x_i)$. Finally, we use the boosting tree algorithms (`Tree`'s oracle) to fit the boosting tree model to the dataset $\{(x_i, r_{i,\texttt{Iso}}^t)\}_{i=1}^n$, acquiring $\texttt{Tree}^t$. This concludes one round of iteration.

- Multiplicative separable model: $\mathcal{F}_{\text{multip}} = \{\texttt{Iso}(p_i) \times \texttt{Tree}(z_i)\}$. The multiplicative separable model shares almost the same procedure, except for the residuals are defined as $r_{i,\texttt{Tree}}^t = y_i / \texttt{Tree}^{t-1}(x_i)$ and $r_{i,\texttt{Iso}}^t = y_i / \texttt{Iso}^t(x_i)$, respectively. In the actual implementation, several modifications are introduced to address arising computational issues such as numerical instability. One technique is to take logarithm and transform the multiplication into addition. We find that this technique significantly improves the stability of Algorithm 1.

Note that for both the additive and multiplicative model, the submodels learned by our Submodel Descent algorithm might be off by a constant. This is because we have multiple solutions that gives the same demand model: $(\texttt{Iso}, \texttt{Tree})$ versus $(\texttt{Iso} + c, \texttt{Tree} - c)$ for the additive separable model, and $(\texttt{Iso}, \texttt{Tree})$ versus $(c\texttt{Iso}, \texttt{Tree}/c)$ for the multiplicative separable model. Nevertheless, this is not an issue since we only care about the final composite, not the individual contribution of a submodel itself.

We now move on to some standard regularity and continuity conditions necessary for our analysis. In the following, all the norms are Euclidean norm.

Definition 2 (Lipschitz smooth and strongly convex). A function $\Phi(v) : V \subseteq \mathbb{R}^d \to \mathbb{R}$ is *L-Lipschitz smooth* for some $L > 0$ if Φ is continuously differentiable and for all $v, v' \in V$, we

have

$$\Phi(v') \le \Phi(v) + \langle \nabla\Phi(v), v' - v\rangle + \frac{L}{2}\|v' - v\|^2. \tag{13}$$

A function $\Phi(v) : V \subseteq \mathbb{R}^d \to \mathbb{R}$ is *μ-strongly convex* for some $\mu > 0$ if Φ is continuously differentiable and for all $v, v' \in V$, we have

$$\Phi(v') \ge \Phi(v) + \langle \nabla\Phi(v), v' - v\rangle + \frac{\mu}{2}\|v' - v\|^2. \tag{14}$$

We now specify the regularity of empirical loss function under the additive and multiplicative models. For additive models, $\hat{\mathcal{L}}_{\text{add}}(\mathbf{g}_1, \cdots, \mathbf{g}_m) = \sum_{i=1}^n (y_i - \sum_{j=1}^m g_{j,i})^2$, which is $(2m)$-smooth (proof is delayed to appendix). While this function is easily shown as convex, it is not μ-strongly convex. An evidence is that strongly convex functions on convex sets can only have one unique minimum point, see Section 9.1.2. in Boyd and Vandenberghe (2004). However, $\hat{\mathcal{L}}_{\text{add}}$ can have multiple minimum points. For example, consider the case where $n = 1$, $m = 2$, and thus $\hat{\mathcal{L}}_{\text{add}} = (y_1 - g_{1,1} - g_{2,1})^2$. Suppose that $(g^*_{1,1}, g^*_{2,1})$ is a minimum point, then $(g^*_{1,1} + c, g^*_{2,1} - c)$ for all $c \in \mathbb{R}$ are also minimum points.

We stress that the regularity conditions presented in Definition 2 serve as criteria for the composition function h and the resulting empirical loss $\hat{\mathcal{L}}$ with respect to $\mathbf{g}$, rather than with respect to x_i. For instance, in the MER's case, we use isotonic regression, and a monotonic function is in general not L-smooth or μ-strongly convex, and even need not to be differentiable or convex. But the submodel class `Iso` and `Tree` are convex sets, and we showed that the composite model used in the MER's case lead to L-smooth and convex empirical loss functions.

Based on above regularities, we present the main theorem of this section. We first state the necessary notations and assumptions. We use $\hat{g}^t$ to denote $(\hat{g}^t_1, \cdots, \hat{g}^t_m)$, i.e., the output of Algorithm 1 after iteration t. Also, we denote by $\hat{g}$ a minimum point of the functional $\hat{\mathcal{L}}$. To ensure the sequence generated by Algorithm 1 is well-defined, we assume that the level set $S = \{\mathbf{g} \in \prod_{j=1}^m \mathcal{G}\,|_{x_1,\cdots,x_n} : \hat{\mathcal{L}}(\mathbf{g}) \le \hat{\mathcal{L}}(\hat{\mathbf{g}}^0)\}$ is compact. As a typical requirement, We additionally assume that $\mathcal{G}_j\,|_{x_1,\cdots,x_n}$ are all convex sets. We let $R = \max_{\mathbf{g}\in S} \|\mathbf{g} - \hat{\mathbf{g}}\|$ be the "radius" of S. To make the theorem crisp, we use the short-handed notation $\hat{g}^t = \{\hat{g}^t_1, \cdots, \hat{g}^t_m\}$ as the collection of submodels acquired at the tth iteration of Algorithm 1 and $\hat{g} = \{\hat{g}_1, \cdots, \hat{g}_m\}$ as the collection of minimizers of problem Eq. (9).

THEOREM 4. *Suppose that exact oracles are applied to Algorithm 1, we have the following results:*

- ***(Smooth and convex)** If $\hat{\mathcal{L}}(\mathbf{g}_1, \cdots, \mathbf{g}_m)$ is smooth and convex, then Algorithm 1 converges:*

$$\lim_{t\to\infty} \hat{\mathcal{L}}(\hat{g}^t) - \hat{\mathcal{L}}(\hat{g}) = 0.$$

Also, every limit point of the sequence $\{\hat{g}^t\}$ is a minimum point of $\hat{\mathcal{L}}$.

- ***(L-smooth and convex)*** *If $\hat{\mathcal{L}}(\mathbf{g}_1, \cdots, \mathbf{g}_m)$ is L-smooth and convex, then Algorithm 1 converges sublinearly:*

$$\hat{\mathcal{L}}(\hat{g}^t) - \hat{\mathcal{L}}(\hat{g}) \leq \frac{c}{\sigma} \cdot \frac{1}{t},$$

where $\sigma = 1/(2LR^2n^2)$ and $c = \max\{4\sigma - 2, \hat{L}(\hat{g}^0) - \hat{L}(\hat{g}), 2\}$.

- ***(L-smooth and μ-strongly convex)*** *If $\hat{\mathcal{L}}(\mathbf{g}_1, \cdots, \mathbf{g}_m)$ is L-smooth and μ-strongly convex, then Algorithm 1 converges linearly:*

$$\hat{\mathcal{L}}(\hat{g}^t) - \hat{\mathcal{L}}(\hat{g}) \leq \left(\frac{mL^2}{mL^2 + \mu^2}\right)^t (\hat{\mathcal{L}}(\hat{g}^0) - \hat{\mathcal{L}}(\hat{g})).$$

The theorem furnishes convergence guarantees and computational complexities for Algorithm 1 under varying assumptions about the regularity of $\hat{\mathcal{L}}$. When considering only the smoothness and convexity, we can anticipate convergence results without being able to guarantee the speed of convergence. This result is based on Tseng (2001) and Bertsekas (1997). By adding the L-smoothness assumption, we can ensure a sublinear convergence rate of $\mathcal{O}(1/t)$, following a similar approach in Hong et al. (2017). Moreover, if we can assume μ-strong convexity, we can achieve an exponential rate of convergence, influenced by the ideas presented in Li et al. (2017). The proof of all three results centers around transforming the functional optimization problem Eq. (9) to a convex optimization problem Eq. (12), and transforming Algorithm 1 to Algorithm 2.

EXAMPLE 1 (CONTINUED). In the MER's case, from Example 1 we know that the used additive separable model generates L-smooth and convex empirical loss function, and therefore enjoys sublinear rate using Algorithm 1. The multiplicative separable model is constrained to produce positive values only, and the corresponding $\hat{\mathcal{L}}$ is also L-smooth and convex, therefore it also enjoys sublinear rate.

Inexact Oracle. Now we replace the exact oracles in Algorithm 1 with inexact ones that find the approximate optimal solution within a certain error margin. Let the maximal error incurred in iteration t be bounded by ε_t, we are interested in the convergence result of the inexact version of Algorithm 1 under the error margin sequence $\{\varepsilon_t\}$. We would like to note that our definition of inexact oracle is different from what has been studied in previous coordinate descent literature, such as Tappenden et al. (2016) and Yang et al. (2019). Previous literature focuses on block coordinate *gradient* descent algorithms, where inexactness was usually defined as the error in gradient steps. Ours focus on the error in the solution towards the coordinate minimization problem (Eqs. 10 and 11). The two kinds of coordinate descent algorithms are based on two different kinds of oracles. This difference renders previous proof techniques ineffective, and constitutes the main difficulty of our analysis. Because our oracle does not provide gradient information but only returns the minimizers at each coordinate, it becomes much more difficult to control how error in previous coordinates would propagate to subsequent coordinates.

For the following theorem, we still apply the same preliminary assumptions as Theorem 4: level set $S = \{\mathbf{g} \in \prod_{j=1}^{m} \mathcal{G} \mid_{x_1,\cdots,x_n} : \hat{\mathcal{L}}(\mathbf{g}) \leq \hat{\mathcal{L}}(\hat{\mathbf{g}}^0)\}$ is compact, and $\mathcal{G}_j \mid_{x_1,\cdots,x_n}$ are all convex sets. Our proof technique relies on the comparison between the exact and inexact versions of Algorithm 1.

THEOREM 5. *Suppose that inexact oracles are applied to Algorithm 1, and let ε_t denote the error margin incurred at each submodel in iteration t, we have the following results:*

- ***(Smooth and convex)*** *Assume that $\hat{\mathcal{L}}(\mathbf{g}_1, \cdots, \mathbf{g}_m)$ is smooth and convex. If the error margin sequence diminishes to zero, i.e., $\lim_{t\to\infty} \varepsilon_t = 0$, then the inexact version of Algorithm 1 still converges:*

$$\lim_{t\to\infty} \hat{\mathcal{L}}(\hat{g}^t) - \hat{\mathcal{L}}(\hat{g}) = 0.$$

Moreover, every limit point of the sequence $\{\hat{g}^t\}$ is a minimizer of $\hat{\mathcal{L}}$.

- ***(L-smooth and μ-strongly convex)*** *Assume that $\hat{\mathcal{L}}(\mathbf{g}_1, \cdots, \mathbf{g}_m)$ is L-smooth and μ-strongly convex. Given a known error margin sequence $\{\varepsilon_t\}$, the following bound holds:*

$$\hat{\mathcal{L}}(\hat{g}^t) - \hat{\mathcal{L}}(\hat{g}) \leq \left(\frac{mL^2}{mL^2+\mu^2}\right)^t (\hat{\mathcal{L}}(\hat{g}^0) - \hat{\mathcal{L}}(\hat{g})) + c\sum_{s=1}^{t} \left(\frac{mL^2}{mL^2+\mu^2}\right)^{t-s} \varepsilon_s,$$

where $c = \max_{\mathbf{g}\in S} ||\nabla \hat{\mathcal{L}}(\mathbf{g})||^2 \cdot ((\mu+L)^m - \mu^m)/\mu^{m-1}L$. This maximum exists since the level set S is compact.

The main takeaway from Theorem 5 is inexact oracles do not significantly affect the convergence guarantees or convergence rate of the submodel descent algorithm, whether under the smooth and convex case or the L-smooth and μ-strongly convex case. In the smooth and convex case, as long as the error margin sequence diminishes over time, the convergence of Algorithm 4 remains unaffected. In the L-smooth and μ-strongly convex case, the convergence bound consists of a primary term identical to that in Theorem 4, plus an additional error term. This convergence guarantee holds for any error margin sequence $\{\varepsilon_t\}$. Notably, the error produced at each iteration t will decrease linearly in subsequent iterations, matching the rate of the primary term. In addition, a corollary of Theorem 5 is that as long as the error margin sequence shrinks linearly, it does not impact the algorithm's linear convergence rate.

REMARK 2. It's worth noting that the case of L-smooth and convex (without strong convexity) is absent from Theorem 5. This omission is due to the essential role of strong convexity in our proof. Our strategy centers around comparing the inexact algorithm to its exact counterpart at each iteration. It requires the controlling the difference between the two by bounding error propagation across submodels. This strategy relies on a crucial lemma, which provide sensitivity of the minimizer at one coordinate with respect to perturbations at other coordinates. However, the lemma does not hold without strong convexity. For further details, including the proof of Theorem 5, please refer to the appendix.

5. Implementation and Results

In this section, we present our demand learning and price optimization results obtained from the MER dataset. The MER dataset comprises sales data for consumer electronics from the company's online retailing platform spanning from 2019 to 2021. Sales data is recorded every two days for each SKU, and there are approximately 2000 SKUs in total. Following the approach outlined in Section 3 and 4, we explore two separable models introduced in Example 1: the additive separable model, denoted as $\texttt{Iso}(p)+\texttt{Tree}(x)$, and the multiplicative separable model, denoted as $\texttt{Iso}(p)\times\texttt{Tree}(x)$. We conduct two sets of experiments to evaluate the performances of the two separable models: a demand learning experiment that compares their demand prediction accuracy to two naive models, and a price optimization experiment that investigates the quality of their derived pricing decisions.

Product grouping. Since each SKU experienced very few price changes and low sales, we first aggregate the products into four groups. The grouping is based on a product's *demand cumulative distribution function* (*demand CDF*) defined as

$$Q(p) = \text{ (total sales when the price is no larger than } p) \text{ / (total sales).}$$

where p is a selected price. $Q(p)$ is designed to quantify the impact of price changes on the probability of selling a single unit. We assess the product's demand CDF by evaluating it at various price points, and the resulting values become the grouping features. We transform $Q(p)$ into the features for clustering by evaluating its values at several fixed price points. It is saying that we use $[Q(p_1), Q(p_2), \cdots, Q(p_k)]$ for some $p_1 \le p_2 \le \cdots \le p_k$ as the clustering features. Subsequently, we employ the K-means clustering algorithm to aggregate these features into four distinct groups. Table 3 presents some statistics of the product groups. Notably, our analysis reveal that these groups strongly correlate with the product's price distribution over its entire life cycle. To streamline our discussion, we will refer to these groups as the low, mid-low, mid-high, and high price groups.

Table 3 Statistics of the four product groups.

Group	Product count	Price range	Average price	Average sales per-cycle
low price	902	[25.0,9099.0]	1563.5	2.33
mid-low price	675	[720.0, 9999.0]	3973.0	4.39
mid-high price	210	[3499.0, 24600.0]	9464.0	2.46
high price	171	[7062.0, 37368.0]	18280.1	0.65

There are several reasons why we group the products based on their demand CDFs rather than their categorical features. First, we expect price to be the dominant factor when a customer makes

purchase decisions. To a customer, products with the same level of price should be perceived similarly, and this leads to similar demand patterns. Second, the design of separable models prefers a dataset where all products have similar price levels. This is because in both additive and multiplicative separable models, a base demand curve is shared among all products in the group, and all the product-specific demand curves would have the same price domain. It is therefore appropriate to separate products with different price levels into different groups. Third, the categorical features we have are rather limited in the MER dataset. For each product, the important categorical features only include brand, color and market. This does not distinguish products well enough, for example, products from the same brand that are geared towards customers with different spending power.

Another comment we want to make is about why we use demand CDFs instead of the product's price distribution over time. The is because demand CDF tells us not only the price distribution of a product across time, but also the portion of this products sold under a given price. In other words, demand CDF conveys not only the price information of a product, but also how popular the product will be in the market under one specific price. This additional information makes clustering be more accurate.

Finally, the decision to use specifically four clusters stems from careful consideration of the bias-variance trade-off problem. Too many clusters would result in insufficient data per group, leading to high variance in our estimates. Conversely, too few clusters would force us to group products with substantially different price models, introducing significant bias. Through our analysis of the MER dataset, we determined that four clusters provides the optimal balance.

Rolling forecast. We assess model performance using a rolling evaluation approach. The dataset spanning three years is divided into two segments. The initial 34 months serve as the training period, while the final 2 months serve as the testing period. We begin at the start of the testing period, simulating a scenario where only training data is available. At the commencement of each cycle, we retrain every model using the complete historical data. Subsequently, each model is provided with price and contextual features for the upcoming cycle, enabling it to make demand predictions. Then, we generate an demand curve for every product, and run a price optimization algorithm to obtain the optimal price. Afterwards, the actual sales volume is unveiled and added to the training dataset as we transition to the next cycle. Model predictions are continually tracked throughout, and at the end of the process, we calculate each model's loss. This process is reiterated until we reach the conclusion of the testing period.

5.1. Demand Learning

To highlight the advantages of separable models, we compare their performance with two naive implementations: the first is the naive isotonic model, denoted as $\texttt{Iso}(p, x)$, where we train an

isotonic model for each SKU individually. The second is the naive boosting tree model. We use Catboost in this paper, which is considered as the powerful boosting tree model when the categorical features take important roles, denoted as $\texttt{Tree}(p, x)$, which involves running a boosting tree model across all the features, including price, in the dataset. These two baseline methods do not have a separable structure. We note that there are other benchmark options, such as the groupwise isotonic model, where we train a single for all the products in a group. However, the groupwise isotonic model would produce worse demand curves than the naive isotonic model since it would be influenced by outliers and dissimilar products.

Given that our separable model was primarily developed in response to the shortcomings of traditional approaches in learning price elasticity, our first comparison focuses on the demand curves. Figure 4 illustrates the demand curves of four selected products, each from one of the four price groups. The top-right subfigure features the same product that we employed to demonstrate the issue of flat demand curves in the introduction (as shown in Figure 2). Among the products in all four groups, it is readily apparent that the curves generated by the isotonic model remain constant, whereas those produced by the Catboost model frequently exhibit upward trends and fluctuations. As it is already shown at the beginning, the main reason for the poorly-shaped curves is most products have rarely gone through price changes, and it is the scarcity of data and price changes that leads to flat curves for the isotonic model. The ability to impose shape constraints on the resulting demand curves is a key advantage of the separable model. Consequently, all curves generated by the separable models strictly exhibit a decreasing trend. This observation is further substantiated by quantifying the proportion of decreasing curves. Table 4 quantifies the percentage of products with decreasing demand curves. For each product, we compare the predicted demand at the product's lowest possible price with that at the highest possible price. A curve is classified as "decreasing" whenever the model predicts lower demand at the higher end of the price compared to the lower end. We acknowledge that this measurement is a simplification and might not capture the fluctuations between the price bounds, which are consistently observed in the boosting tree model, as evident from the orange curves in Figure 4. Nonetheless, we have found that the two naive models produce a significant proportion of demand curves that are not decreasing, failing their meaningful applications in price optimization.

Table 4 Percentage of products with a decreasing demand curve.

Model	Low	Mid-low	Mid-high	High
Isotonic	35%	39%	38%	46%
Catboost	7%	83%	64%	35%
Additive	100%	100%	100%	100%
Multiplicative	100%	100%	100%	100%

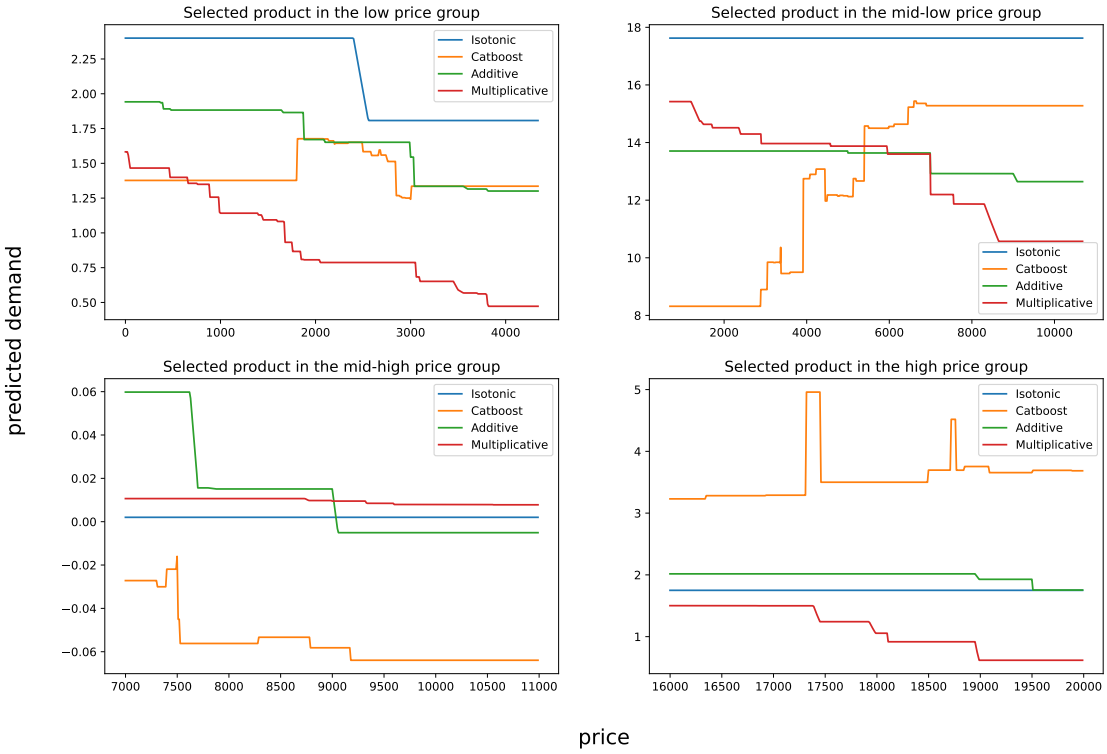


Figure 4 Demand curve derived by the four models.

As our separable model can produce more favorable demand curves, we aim to compare the prediction accuracy of the four models. We evaluate accuracy using three metrics: mean absolute error (MAE), mean squared error (MSE), and coefficient of determination (r^2). Table 5 summarizes the error metrics for all four product groups. An immediate observation is the underperformance of the naive isotonic regression, which exhibits a substantial performance gap compared to the other three models across all three metrics and in all groups. This underscores the clear inadequacy of naive isotonic regression approaches in the context of MER. Furthermore, the performance of the remaining three models is consistently close to each other, with the Catboost model often achieving slightly superior results compared to the two separable models. The additive separable model frequently outperforms the multiplicative separable model and occasionally even outpaces the Catboost model. Consequently, the separable model provides significantly improved price elasticity estimations without substantially compromising overall accuracy.

Another aspect we wish to emphasize is the scalability of the multiplicative separable model. Within the same product group, there may exist a significant variation in the magnitude of sales volumes across different products. While the majority of products yield only single-digit unit sales per cycle, a few products achieve sales in the tens or even hundreds of units per cycle. As previously

Table 5 Loss metrics for the demand prediction of the models.

(a) Low price

Model	MAE	MSE	r^2
Isotonic	1.99	38.72	0.29
Catboost	0.76	16.25	0.70
Additive	0.78	16.81	0.69
Multiplicative	0.82	23.66	0.57

(b) Mid-low price

Model	MAE	MSE	r^2
Isotonic	3.73	62.99	0.17
Catboost	1.15	22.08	0.71
Additive	1.17	21.95	0.71
Multiplicative	1.22	34.92	0.54

(c) Mid-high price

Model	MAE	MSE	r^2
Isotonic	1.74	25.91	0.62
Catboost	1.22	20.24	0.71
Additive	1.22	19.02	0.72
Multiplicative	1.51	53.58	0.22

(d) High price

Model	MAE	MSE	r^2
Isotonic	0.75	2.31	0.06
Catboost	0.38	1.02	0.59
Additive	0.40	1.04	0.58
Multiplicative	0.38	2.19	0.11

discussed, the demand curve generated by the additive separable model fundamentally consists of a base curve from `Iso` plus a constant from `Tree`. This implies that `Iso` learns the absolute change in demand and adapts well to products with smaller-scale sales volumes. Consequently, demand predictions from the additive model tend to be relatively insensitive to price changes for products with high sales volume. In contrast, the multiplicative separable model multiplies the constant from `Tree` by the base curve. This means that `Iso` captures the relative change in the demand curve, rather than the absolute change. As a result, the multiplicative model is better suited to accommodate products with varying sales volume magnitudes. Undoubtedly, this feat comes at a price. We will demonstrate in Section 5.2 that demand curves from multiplicative separable models are more sensitive to price changes, and therefore could potentially produce overly optimistic revenue forecasts.

In summary, both the additive and multiplicative separable models ensure decreasing curves with minimal sacrifice in accuracy when compared to the Catboost model. The additive model, overall, exhibits slightly higher accuracy compared to the multiplicative model, but the latter excels in scalability and proves ideal for datasets with products of varying sales volume magnitudes.

5.2. Price Optimization

After the demand learning phase, demand model is integrated into a price optimization algorithm, presenting two primary challenges. First, the resulting demand curve could be nonlinear. For demonstration, we choose `Iso` as the price submodel, which gives a piecewise linear demand curve, suggesting efficient algorithms such as quadratic optimization methods. However, other submodels except from `Iso` may not necessarily end in well-behaved optimization problems. Complicating matters, MER typically selects prices ending in 49 or 99. This leads to discretization over the

price decision space. In addition, MER would like to retain the flexibility of the price optimization module, ensuring its compatibility with any demand learning model architecture.

Second, the constraints proposed by MER prevent optimization at an individual product level. Figure 5 gives a full list of implemented constraints, which fall into two categories: productwise and cross-product constraints. Productwise constraints are tailored to individual products, such as constraints that ensure price stability over time and alignment with competitors' pricing. Crosswise constraints affect multiple products in the same group, such as constraints that ensure newer generation products are not priced lower than older ones within the same product line. Crosswise constraints require us to conduct one large price optimization altogether, rather than optimize for each product separately.



Figure 5 Constraints of the price optimization.

In light of these challenges, we formulate the price optimization as a mixed integer programming problem. For each product, we discretize the price space, selecting 20 feasible prices. The objective is to maximize the predicted revenue on the given demand model, adhering to the specific constraints set by MER. In this phase, we only optimize for products in their early life cycles, as these products are available on markets and hold the most interest from MER executives. This results in a selection of approximately 150 products from each of the low and mid-low price groups, and 50 products from each of the two high price groups. Despite targeting these relatively popular products, the data scarcity issue still remains. On average, more than half of the time periods recorded no sales for each product.

We run the test in a rolling manner for each group for a span of 10 periods. In each period, we track three variables: the actual revenue, predicted revenue (the revenue forecast at the product's actual price), and optimized revenue (the revenue forecast at the product's optimal price). We aggregate the revenues across all the products for each period. Figure 6 illustrates the revenue

trends over these 10 periods. We also measure a model's predictive performance by computing the mean absolute error (MAE) and the weighted mean percentage error (wMAPE) between the predicted total revenue and the actual total revenue. The optimization performance is described by the percentage revenue increase of optimized total revenue from predicted total revenue. Table 6 summarizes these three metrics for each model and product group.

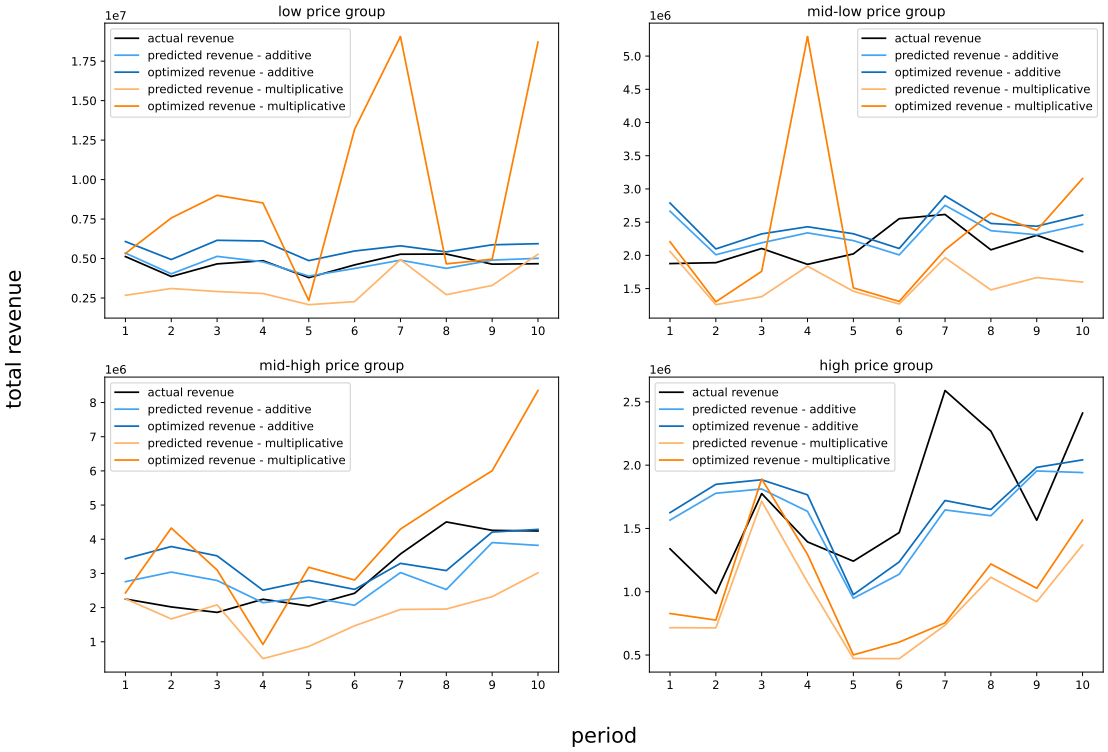


Figure 6 **Total revenues derived by the two separable models.**

Table 6 **Metrics for the price optimization of the separable models.**

Group	Model	MAE	wMAPE	Revenue increase
low price	Additive	3.06×10^6	14%	5%
	Multiplicative	5.76×10^6	27%	48%
mid-low price	Additive	3.14×10^6	7%	21%
	Multiplicative	1.59×10^7	34%	191%
mid-high price	Additive	6.47×10^6	22%	18%
	Multiplicative	1.18×10^7	40%	124%
high price	Additive	4.39×10^6	26%	4%
	Multiplicative	7.73×10^6	45%	12%

Overall, the additive separable model achieves a higher prediction accuracy than the multiplicative separable model. In particular, the additive model excels in the two low price groups, where all the products share similar pricing ranges and magnitudes of sales volume. These findings corroborate earlier insights from the demand learning phase. Moreover, the additive model consistently delivers stable predictions across time. However, both models experienced a performance drop in the two high price groups. This decline is likely due to higher price variance in these two groups, and the products are targeted at a smaller market of customers with higher spending power, which could give rise to noisier data.

A noteworthy observation is that the multiplicative separable model tends to forecast more substantial revenue increases. As depicted in Figure 6, its predictions show greater fluctuation, including occasional spikes. This is because the multiplicative model comes up with more sensitive demand curves compared to the additive model. Tracing back to its roots, in the additive model, the derivative of an demand curve remains the same (i.e. $\texttt{Iso}'(p)$) for all products, irrespective of the sales volume magnitude. Because a large portion of products have single-digit units of sales, $\texttt{Iso}'(p)$ is kept small. Consequently, the additive model predicts only minor demand shifts for each product. On the other hand, in the multiplicative model, the demand curve's derivative is given by $\texttt{Iso}'(p) \times \texttt{Tree}(x)$. Therefore, for higher-demand products, this scaling factor makes the demand curve more sensitive to price changes, leading to bolder and sometimes even doubled optimized revenue prediction. This aligns with our previous observation that `Iso` in the additive model reflects absolute demand changes, while in the multiplicative model it learns the relative demand changes.

6. Concluding Remarks

Together with a consumer electronics retailer in the Middle East (referred to as MER), we consider the problem of context-based demand learning and price optimization with data scarcity. The issue of data scarcity arises due to a limited number of price changes and low sales units for a significant portion of the products. This scarcity of data renders traditional machine learning approaches ineffective in accurately capturing price elasticity. To tackle this challenge, we advocate the adoption of a separable model. This model offers an effective fusion of isotonic regression, which ensures a non-increasing price elasticity, and decision-forest-based models, adept at processing high-dimensional contextual information. We introduce a comprehensive theoretical framework for separable models, conduct an in-depth analysis of the statistical performance of the empirical risk minimizer, and develop a computationally efficient algorithm for identifying the minimizer. Furthermore, we showcase the practical and empirical performance of the separable model using the MER's dataset, with a particular emphasis on the realm of price elasticity.

The merits of separable model can be summarized from three perspectives. From a modeling perspective, the separable model may excel in striking a harmonious balance between predictive

power and model interpretability. Its inherent flexibility empowers practitioners to craft tailored models that harness structural nuances and prior knowledge within different data segments. From a theoretical perspective, separable model enjoys competitive statistical guarantees and efficient computational accessibility. The theorems show that it yields a lower estimation error when compared to non-separable models, all while maintaining a computationally tractable empirical risk minimization framework. From an applicative perspective, the separable model provides an effective remedy for addressing the data scarcity and rising demand curve issue, as exemplified in the MER's case. Our empirical findings underscore that separable models yield informative demand curves, all without significant trade-offs in prediction accuracy.

Currently, our demand learning and pricing adheres to the conventional estimation-then-optimization framework (see, e.g., Bertsimas and Kallus 2020, Lei et al. 2023). Recent literature suggests the direct mapping of data to decision-making processes. Notable frameworks in this vein include the Operational Data Analytics (ODA) framework (see, e.g., Feng and Shanthikumar 2022 and Feng and Shanthikumar 2023) and the integrated-estimation-optimization framework (see, e.g., Elmachtoub and Grigas 2022 and Elmachtoub et al. 2023b). It is our keen interest to explore the integration of the separable model structure into such frameworks, which could potentially revolutionize how we leverage separable models for decision-making. Furthermore, we aim to extend the application of the separable model to diverse operational challenges, encompassing areas like manufacturing, assortment optimization, and inventory control.

The collaboration with MER, as presented in this paper, has sparked inspiration for several subsequent works. We are pleased to highlight that our research has motivated Bu et al. (2022b) to explore online dynamic pricing problems with separable models. Furthermore, the challenges associated with poor elasticity estimation in purely nonparametric models, identified in both Bu et al. (2022b) and our paper, have driven the adoption of a semi-parametric partially linear model for demand representation in Liu et al. (2023).

7. Acknowledgements

The authors sincerely thank the department editor, associate editor, and the three anonymous reviewers for their insightful comments and constructive suggestions, which have significantly improved the quality of this work. The authors also acknowledge the support of the MIT Data Science Laboratory. Menglong Li was partially supported by the Hong Kong Research Grants Council (RGC) General Research Fund [Grant CityU11508223].

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Online Appendix for "Contextual Offline Demand Learning and Pricing with Separable Models"

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In the appendix we present supplementary empirical results and proofs of major theorems in the paper.

EC.1. Cross Validation for Hyperparameter Tuning

In Section 2, we mentioned that we focus on the scenario where feature segmentation, the choice of composition function, and the selection of submodel classes are predetermined. While in the MER's case they are decided based on domain knowledge and optimization objectives, it is also an interesting direction to explore the possibility of using cross validation methods to tune these hyperparameters.

To investigate this possibility, we designed a simulation study to assess whether cross-validation methods can effectively identify optimal feature segmentation points. For each round of experiment, we construct a synthetic dataset comprising 30 features and 1000 data points. The true data generating model is $y = g_1(x^{(1)}) + g_2(x^{(2)}) + \eta$, where the feature vector $x = (x^{(1)}, x^{(2)})$ contains a natural separation at d^*_{sep}th feature, representing the optimal segmentation point. This true segmentation point remains hidden from the analysis process, simulating real-world scenarios where optimal feature groupings are unknown. To detect these underlying feature segments, we implemented a grid search algorithm utilizing cross validation. We use the additive separable model

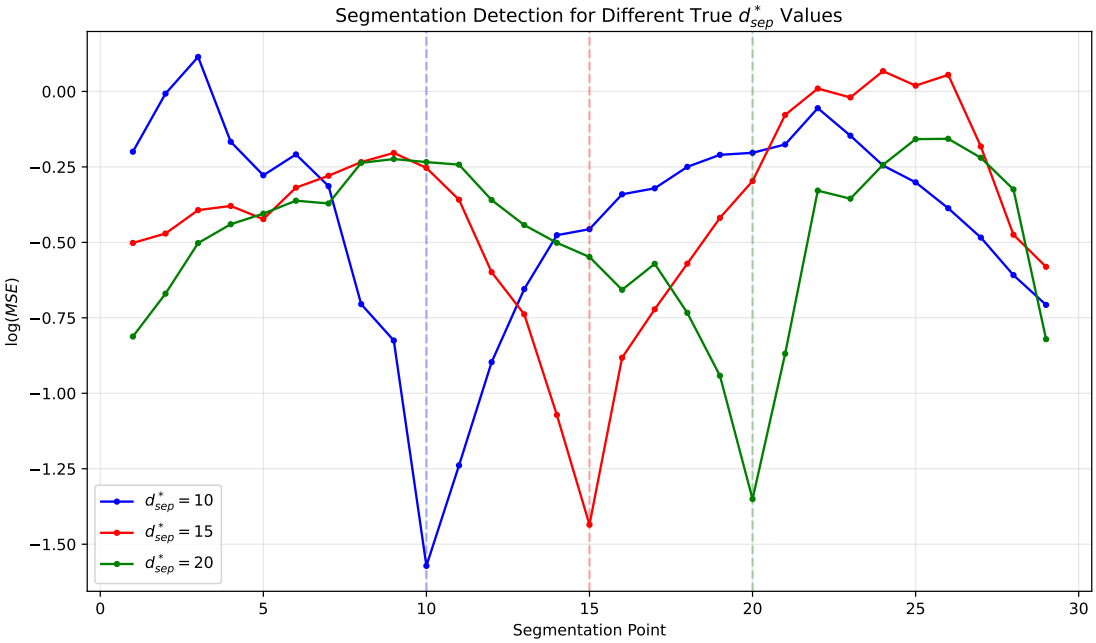


Figure EC.1 Cross validation methods to search for the optimal feature segmentation.

$g_1(x_{1:d_{sep}}) + g_2(x_{d_{sep}:30})$, where g_1 and g_2 are both three-layered fully connected neural networks. The search evaluates all possible separation points d_{sep}, systematically testing divisions from feature 1 through feature 30. The performance of each separation point is evaluated using the classical k-fold cross validation method, with $k = 5$.

Our simulation experiment yields evidences regarding the effectiveness of cross-validation for detecting optimal feature segmentation points. The results are visualized in Figure EC.1, which displays the logarithmic Mean Squared Error (log(MSE)) across different separation points for three distinct true d^*_{sep} values (10, 15, and 20). For each tested scenario, we observe a distinctive V-shaped error pattern, with the minimum error consistently occurring at the true d^*_{sep} value. We also observe that the model performance has some tolerance on feature segmentation, as slight deviation from the true ones would still performs relatively well. However, the error is still very sensitive to hyperparameter, as evidenced by the steep slopes in error around the true segmentation. These findings strongly support the viability of using cross-validation for automated feature segmentation detection.

EC.2. Bias-variance Trade-off

One of the challenges of the separable model framework is the true data generating model might not be separable, raising the concern on the risk of mis-specification. Bias-variance trade-off provides theoretical justification for the separable model framework. In short, the model mis-specification risk corresponds to the bias term, and the sample complexity corresponds to the variance term.

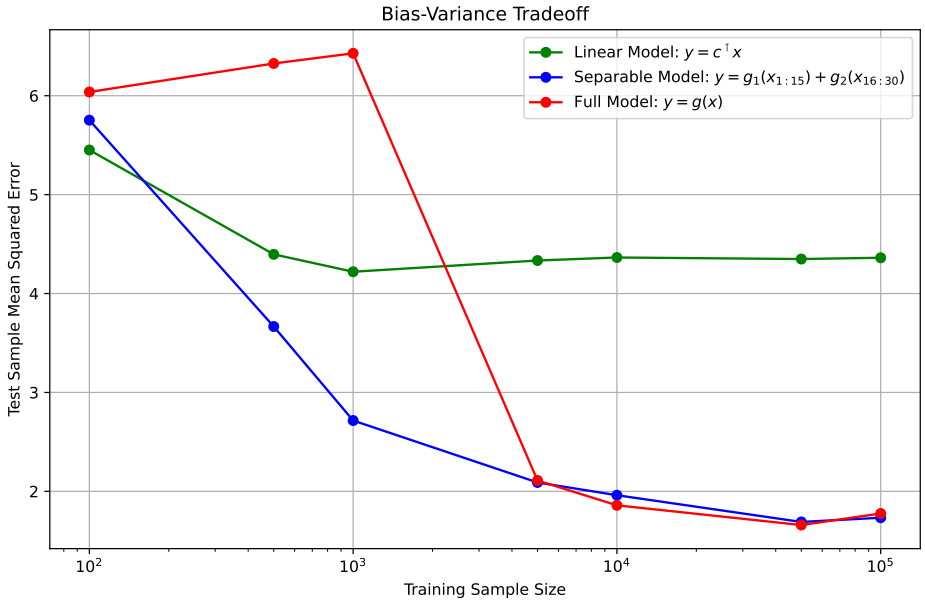


Figure EC.2 Evaluation of the generalizability of models trained on different dataset sizes.

In the MER's case, the demand learning process has the problem of data scarcity, which suggests that current data volume may be insufficient to reliably train highly complex models. Therefore, we need to add structures to our model, taking the risk of model mis-specification to trade for increased statistical performance.

To demonstrate the merit of separable model under limited data, we run a simulation experiment on a synthetic dataset. We construct a complex nonlinear, nonseparable data generating process with 30 input features to evaluate the bias-variance trade-off between different model architectures. The true underlying function incorporates multiple forms of nonlinearity and feature interactions, deliberately structured to deviate from the additive separable form. The data generating process combines six distinct types of nonlinear relationships: periodic interactions, polynomial terms, exponential and logarithmic transformations, threshold effects, pairwise interactions, and higher-order interactions that involves more than three features. We evaluated three models of increasing complexity: (1) a linear model ($y = c^\intercal x$) that assumes simple additive effects, (2) an additive separable model ($y = g_1(x^{(1)}) + g_2(x^{(2)})$) that allows for nonlinear but independent feature effects, and (3) a complex neural network ($y = g(x)$) that takes all input features. The linear model serves as a high-bias, low-variance baseline and the neural network represents a low-bias, high-variance approach capable of approximating the true generating process given sufficient data.

The experiment is run as follows. We train each model with different volume of data, and then test its predictive power on a test dataset with 1000 data points. The mean squared error incurred

on the test dataset demonstrates the generalization ability of the trained model. The result is shown in Figure EC.2. As we can see, the linear model has a stable but suboptimal performance. Its error quickly plateaus even as the sample size increases, suggesting its high bias prevents further improvement in large dataset sizes. The full model shows interesting dynamics. With small sample sizes (below $1,000$), it performs poorer than the linear model, demonstrating high variance due to overfitting. However, once the sample size reaches approximately $10,000$, its performance improves dramatically, eventually achieving comparable results to the separable model at large sample sizes. Finally, we see that the separable model presents a compelling compromise. At small sample sizes ($\sim 1,000$), it significantly outperforms the full model while maintaining better accuracy than the linear model. It shows consistent improvement as the sample size increases. Most notably, it provides stable and competitive performance across all sample sizes, making it particularly valuable when training data is limited.

These results empirically support our theoretical analysis: while the full model can potentially capture all relationships in the data, it requires substantial training data to realize this potential. The separable model, by introducing beneficial structural constraints, achieves a better balance between model expressiveness and sample efficiency, particularly in the practical regime of limited data availability. It achieves balance in the bias-variance trade-off problem, becoming our choice for the demand learning task for MER.

EC.3. Demand Forecasting and Pricing with Synthetic Product Data

While our analysis in Section EC.2 demonstrates the bias-variance tradeoff of different model architectures, a more crucial practical consideration is how these models perform when used for pricing decisions. Even when demand estimation errors are comparable between models, structural properties—particularly the monotonicity guarantees for price effects—can lead to significantly different revenue outcomes. This appendix extends our investigation to address this question directly through experiments with synthetic data.

For our experimental setup, we generate synthetic product demand data with varying contextual features, decreasing price sensitivity, non-linear effects, and feature interactions that make the true model non-separable. The synthetic dataset exhibit data scarcity issue: each product have 10 features, we have 50 products in total, where for each product we only have 10 data points. We then compare three modeling approaches: a naive productwise isotonic regression that fits separate models for each product; our proposed separable model that combines isotonic regression for price with CatBoost for contextual features; and a complex model using CatBoost to model all features together.

Table EC.1 presents the key metrics from our experiments, showing that the models achieve comparable prediction accuracy (as measured by MAE, MSE, and R^2), but have different structural properties and revenue implications.

Table EC.1 Model performance under synthetic product data.

Model	MAE	MSE	R^2	Decreasing Curves (%)	Predicted Revenue	Actual Revenue
Isotonic	2.945	35.19	0.423	72.92	1705	970.6
CatBoost	3.386	41.50	0.319	95.83	1821	1181
Additive	3.675	40.48	0.336	100.0	1779	1288

To better understand these differences, we examine the predicted demand curves for specific products. Figure EC.3 shows demand curves for four randomly selected products. We also have the corresponding revenue curves in Figure EC.4 that illustrates the consequences of these demand predictions for pricing decisions. As we see, Isotonic regression obtains flat price elasticity curves, failing to capture the elasticity of products; Catboost would occasionally generate spikes in its demand curves. The low quality of demand curves lead to suboptimal pricing decisions. We plotted the revenue curves from each model, and it can be seen that the high quality of demand curves lead to stable pricing decisions, which results in the dominant performance of the separable model.

It is also worth noting that the separable model dominates when we have moderate data scarcity. When we increase the number of observed data points for each product, the performance of a fully complex model quickly catch up and sometimes even surpasses the separable model. This can be explained by the bias-variance trade-off perspective: when we have abundant data, the bias term would dominate, and a more flexible model would have less bias and therefore more accurate. We observe the same phenomena as when the ratio $(\#\text{ data points})/(\#\text{ products} \times \#\text{ features}) > 20$, the Catboost model actually consistently performs the best.

These findings reinforce our theoretical analysis of the bias-variance tradeoff and provide strong empirical support for our modeling approach. The separable model provides a robust framework for pricing optimization by enforcing critical domain knowledge (decreasing demand with price) while still capturing complex patterns in contextual features. This makes it particularly well-suited for real-world pricing applications where reliability and consistency are as important as predictive accuracy.

EC.4. Submodel Descent - Selection Rules

In each iteration of the Submodel Descent algorithm, we consider the selection rule where we cyclically optimize each submodel exactly once. However, in actual implementation this might not be the most efficient way of training. For example, in MER's case, the isotonic regression submodel for price is faster to train than the Catboost submodel for contextual features. This

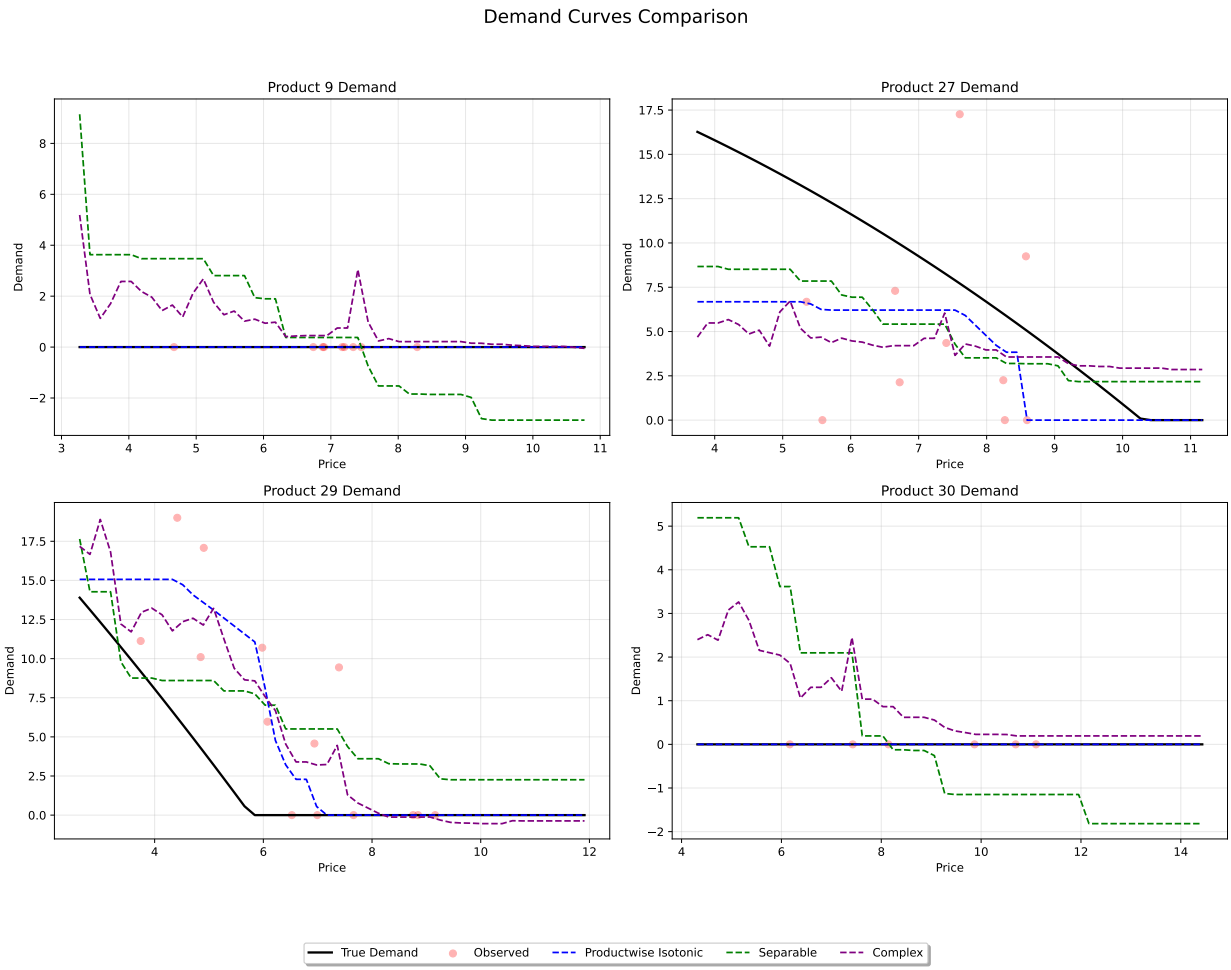


Figure EC.3 Predicted demand curves for four representative products.

suggests the opportunity where one can improve the computational runtime by carefully choosing the selection rule of submodels within each iteration. According to the literature on coordinate descent algorithms, there are four major selection rules:

- Cyclic: update the submodels in a fixed order.
- Randomized sampling: randomly select a submodel following some fixed distribution.
- Greedy: select the submodel such that the objective function is minimized most or the gradient has the largest size.
- Essentially cyclic: updates each submodel at least once within a fixed window.

When some submodels converge much faster than others (let's call them small submodels and large submodels respectively), it is natural to consider the following selection rules to improve the efficiency of the algorithm:

- Randomized sampling with a distribution that assigns higher probability to the small submodels.

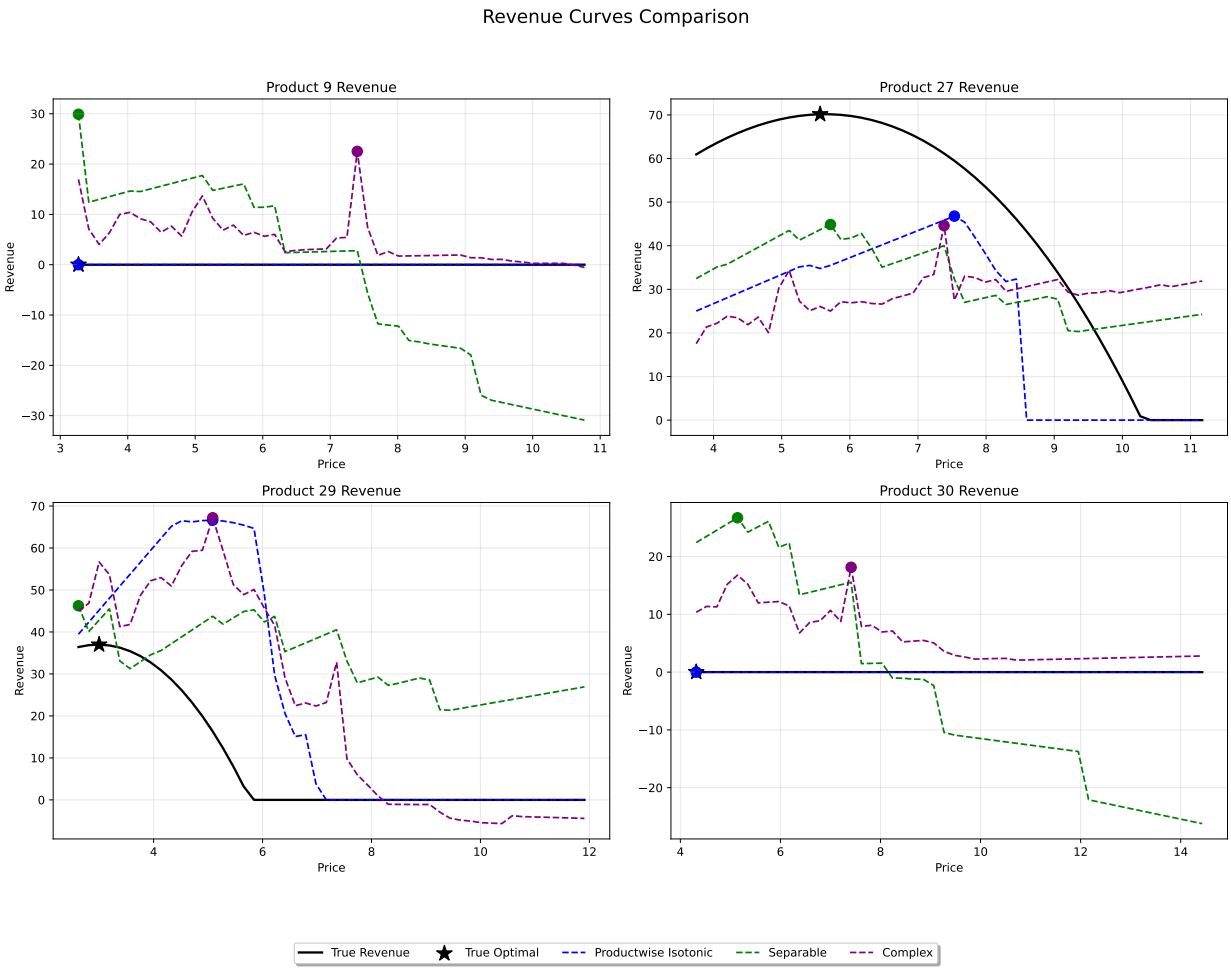


Figure EC.4 Revenue curves derived from the demand predictions. Black stars indicate the optimal price points for the true demand function. Colored circles indicate the optimal price points for each model.

- Essentially cyclic rule that only update large submodel once within a long fixed window.

We conducted experiments to evaluate different coordinate selection rules using a synthetic real estate pricing dataset comprising 30 features and 1000 data points. To effectively test these selection strategies, we implemented an additive separable model with three distinct submodels, each processing 10 features. The submodels consisted of a CatBoost regressor, a decision tree, and a linear regression model, with CatBoost requiring significantly more computational time than the other two models. We implemented and compared three coordinate selection rules: cyclic, which updates submodels in a fixed sequence; randomized, where we assigned a lower sampling probability to the computationally intensive CatBoost model; and essentially cyclic selection, where we updated the CatBoost model once every two iterations while maintaining regular updates for other models.

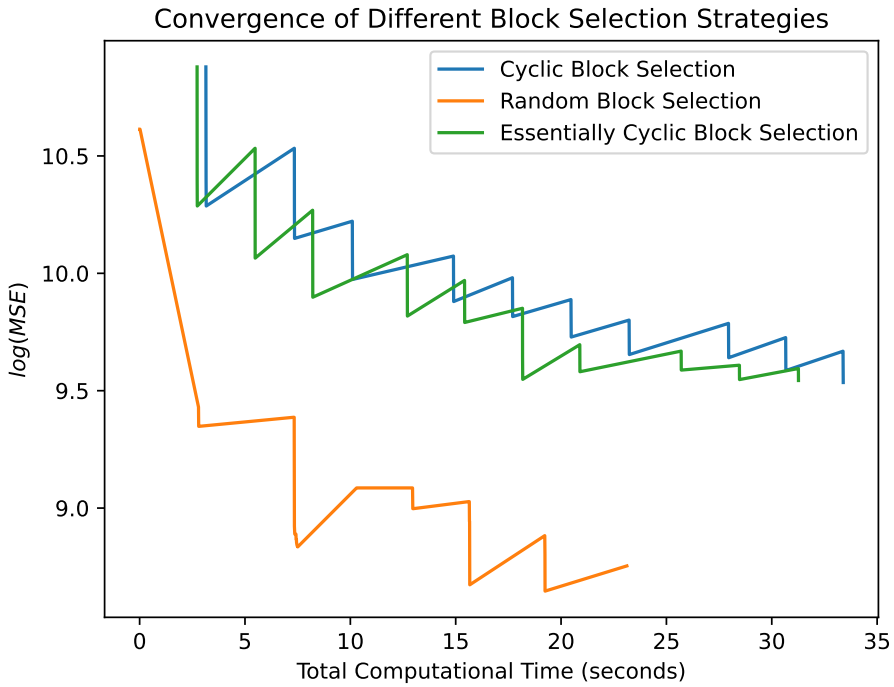


Figure EC.5 Runtime of the three selection rules on a synthetic dataset.

The results, as displayed in Figure EC.5, reveal several key insights about the convergence behavior and computational efficiency of each approach: The randomized selection rule demonstrates superior performance, achieving lower Mean Squared Error (MSE) values more rapidly than the other approaches. This faster convergence can be attributed to the strategic allocation of computational resources, where the algorithm more frequently updates the simpler models (decision tree and linear regression) while selectively updating the computationally intensive CatBoost model. The sharp initial decrease in MSE suggests that this approach quickly captures the main patterns in the data. The cyclic and essentially cyclic selection rules show similar convergence patterns, though the essentially cyclic approach maintains a slightly lower MSE throughout most of the runtime. The saw-tooth pattern observed in both methods corresponds to the periodic updates of the CatBoost model, with temporary increases in error following updates to simpler models and significant improvements after CatBoost updates. These findings suggest that incorporating model complexity into the selection strategy can significantly improve the efficiency of coordinate descent methods. The randomized selection rule's superior performance indicates that balancing update frequency based on computational cost can lead to faster convergence while maintaining high accuracy. This approach appears particularly valuable in scenarios where submodels have substantially different computational requirements.

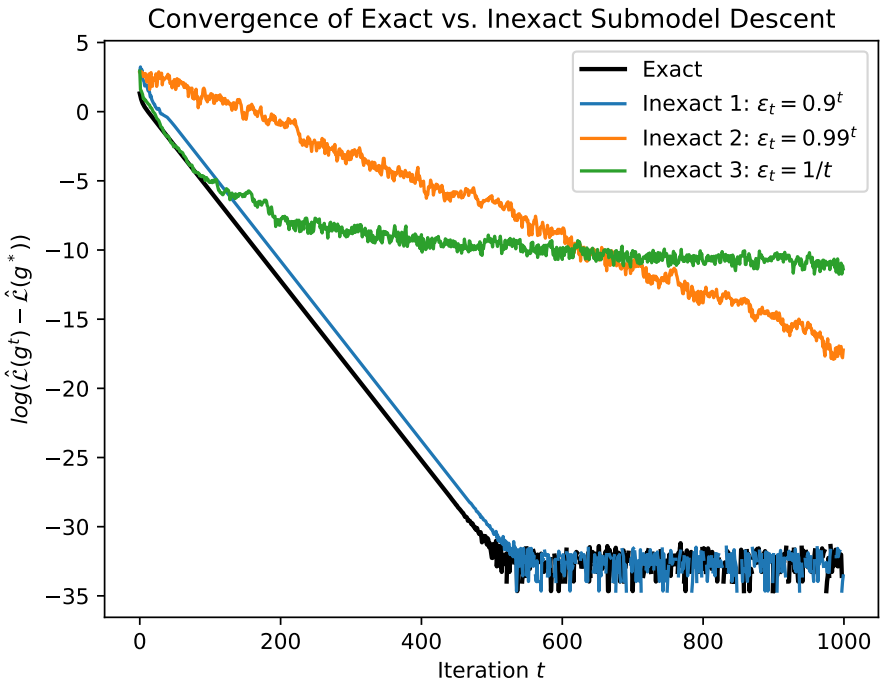


Figure EC.6 **Exact vs. inexact oracles on a quadratic programming problem.**

EC.5. Submodel Descent - Exact vs. Inexact Oracles

To support our convergence guarantees of the Submodel Descent algorithm, we run submodel descent algorithms on a hypothetical quadratic objective function $\mathcal{L}(\mathbf{g}) = \mathbf{g}^\intercal \mathbf{A}\mathbf{g}/2 + \mathbf{b}^\intercal \mathbf{g}$. This objective function is L-smooth and μ-strongly convex, as we let $\mathbf{A}$ to be positive definite. We randomly generate the objective function, and run four versions of submodel descent algorithms on the same objective. The first algorithm is the exact version, and the other three versions are inexact ones with different error margin sequences $\{\varepsilon_t\}$. The result is shown in Figure EC.6.

The simulation results align closely with the upper bounds established in Theorem 4 and Theorem 5, providing several key insights:

- Exact algorithm: The exact version demonstrates linear convergence, consistent with the theoretical guarantee of Theorem 4.
- Inexact algorithms: As stated in Theorem 5, the convergence rate of the inexact oracle is dominated by the slower factor between the convergence rate of the exact version, and the accumulated influence of errors in current and previous iterations. In inexact version 1, where the error margin shrinks faster than the convergence rate of the exact algorithm, the convergence is ultimately determined by the rate of the exact version. In inexact version 2, where the error margin shrinks more slowly, the accumulated error dominates, leading to a slower convergence rate. Nonetheless, the convergence remains linear, confirming the theoretical assertion that the influence of errors is

bounded on the order of $O(\varepsilon)$. Finally, in inexact version 3, where the error margin decreases sublinearly ($\varepsilon_t \sim 1/t$), the algorithm's convergence rate is also sublinear. This result further reinforces the observation that the error's influence is $O(\varepsilon)$.

Overall, the simulation results matches the theoretical predictions, providing strong empirical support for both Theorem 4 and Theorem 5.

EC.6. Proof of Proposition 1

Let $\mathcal{C}_j$ be the smallest (δ/mL)-cover of $\mathcal{G}_j$. Consider the set $\mathcal{C} = \{h(g_1, \cdots, g_m) \mid g_j \in \mathcal{C}_j\}$, which has $\prod_{j=1}^m \mathcal{N}(\mathcal{G}_j, L^2(\mathbb{P}_n), \delta/mL)$ elements. For brevity, denote the multi-valued function $(g_1, \cdots, g_m)$ by $\mathbf{g}$. We now prove that $\mathcal{C}$ is a δ-cover of $\mathcal{F}$.

For every $f, f' \in \mathcal{F}$, we have

$$\begin{aligned}
\|f - f'\|_{L^2(\mathbb{P}_n)} &= \left[\frac{1}{n}\sum_{i=1}^n \left(h\left(\mathbf{g}(x_i)\right) - h\left(\mathbf{g}'(x_i)\right)\right)^2\right]^{1/2} \\
&\le \left[\frac{1}{n}\sum_{i=1}^n L^2 \|\mathbf{g}(x_i) - \mathbf{g}'(x_i)\|_2^2\right]^{1/2} \\
&= L\left[\frac{1}{n}\sum_{i=1}^n \sum_{j=1}^m \left(g_j(x_i) - g_j'(x_i)\right)^2\right]^{1/2} \\
&= L\left[\sum_{j=1}^m \|g_j - g_j'\|_{L^2(\mathbb{P}_n)}^2\right]^{1/2} \\
&\le L\sum_{j=1}^m \|g_j - g_j'\|_{L^2(\mathbb{P}_n)}.
\end{aligned}$$

where the first inequality follows from the Lipschitz continuity of h, and the second inequality follows from the L^1-L^2 norm inequality. Now, for every $f \in \mathcal{F}$, for its jth submodel g_j, we can find a submodel $g_j' \in \mathcal{C}_j$ such that $\|g_j - g_j'\|_{L^2(\mathbb{P}_n)} \le \delta/mL$. Notice we have $f' = h(g_1', \cdots, g_m') \in \mathcal{C}$ and $\|f - f'\|_{L^2(\mathbb{P}_n)} \le L\sum_{j=1}^m \|g_j - g_j'\|_{L^2(\mathbb{P}_n)} \le \delta$. This implies $\mathcal{C}$ is a valid δ-cover of $\mathcal{F}$, which leads to $\mathcal{N}(\mathcal{F}, L^2(\mathbb{P}_n), \delta) \le |\,\mathcal{C}\,| = \prod_{j=1}^m \mathcal{N}(\mathcal{G}_j, L^2(\mathbb{P}_n), \delta/mL)$. Taking logarithm on both sides would then give the desired result. □

EC.7. Proof of Theorem 1

We recall that $\|a\|_{L^2(\mathbb{P}_n)}^2 = \frac{1}{n}\sum_{i=1}^n a_i^2$. Because $\hat{f}_n$ is the empirical risk minimizer that satistfies

$$\hat{f}_n \in \arg\min_{f\in\mathcal{F}} \frac{1}{n}\sum_{i=1}^n (f(x_i) - y_i)^2 = \arg\min_{f\in\mathcal{F}} \|f - y\|_{L^2(\mathbb{P}_n)}^2,$$

we have $\|\hat{f}_n - y\|_{L^2(\mathbb{P}_n)}^2 \le \|f^* - y\|_{L^2(\mathbb{P}_n)}^2$. This leads to

$$\begin{aligned}
\|\hat{f}_n - f^* + f^* - y\|_{L^2(\mathbb{P}_n)}^2 &= \|\hat{f}_n - f^*\|_{L^2(\mathbb{P}_n)}^2 + 2\langle \hat{f}_n - f^*, f^* - y\rangle_{L^2(\mathbb{P}_n)} + \|f^* - y\|_{L^2(\mathbb{P}_n)}^2 \\
&\le \|f^* - y\|_{L^2(\mathbb{P}_n)}^2.
\end{aligned}$$

Here, the inner product is defined as $\langle a,b\rangle_{L^2(\mathbb{P}_n)}=\frac{1}{n}\sum_{i=1}^n a_ib_i$. Recall that we can view the well-specified case from the conditional expectation-noise perspective that $y_i=f^*(x_i)+\eta_i$, therefore the above inequality gives us $\|\hat{f}_n-f^*\|^2_{L^2(\mathbb{P}_n)}\le 2\langle\eta,\hat{f}_n-f^*\rangle_{L^2(\mathbb{P}_n)}$, which is called the basic inequality.

The basic inequality gives us

$$\|\hat{f}_n-f^*\|^2_{L^2(\mathbb{P}_n)}\le 4\langle\eta,\hat{f}_n-f^*\rangle_{L^2(\mathbb{P}_n)}-\|\hat{f}_n-f^*\|^2_{L^2(\mathbb{P}_n)}.$$

Notice that since x_is are fixed, the ERM is essentially decided by the noises η_i. Taking expectations on both sides of the inequality, we have

$$\mathbb{E}_\eta\|\hat{f}_n-f^*\|^2_{L^2(\mathbb{P}_n)}\le\mathbb{E}_\eta\sup_{\bar{f}\in\mathcal{F}-f^*}\frac{1}{n}\sum_{i=1}^n 4\eta_i\bar{f}(x_i)-\bar{f}(x_i)^2.$$

Here, $\mathcal{F}-f^*$ is the model class $\mathcal{F}$ re-centered at f^*, i.e., $\mathcal{F}-f^*=\{f-f^*\mid f\in\mathcal{F}\}$. The right hand side is called the *offset sub-Gaussian complexity*. Liang et al. (2015) introduced a crucial lemma that bounds the offset sub-Gaussian complexity, which is a key ingredient of our result.

LEMMA EC.1 **(Liang et al. (2015), Lemma 6)**. *For any fixed $x_1,\cdots,x_n$ and any $C>0$, if $\eta_i, i=1,\cdots,n$ are 1-sub-Guassian, we have:*

$$\mathbb{E}_\eta\sup_{\bar{f}\in\mathcal{F}}\left[\frac{1}{n}\sum_{i=1}^n\eta_i\bar{f}(x_i)-C\bar{f}(x_i)^2\right]\le\inf_{\delta\ge0,\alpha\in[0,\delta]}\left\{\frac{(2/C)\log\mathcal{N}(\mathcal{F},L^2(\mathbb{P}_n),\delta)}{n}+4\alpha+\frac{12}{\sqrt{n}}\int_\alpha^\delta\sqrt{\log\mathcal{N}(\mathcal{F},L^2(\mathbb{P}_n),\epsilon)}d\epsilon\right\}. \tag{EC.1}$$

We are dealing with σ^2-sub-Gaussian noises, so we take $C=1/4\sigma$ and $\alpha=0$ and multiplying both side by 4σ would give us

$$\begin{aligned}\mathbb{E}_\eta\|\hat{f}_n-f^*\|^2_{L^2(\mathbb{P}_n)}&\le 4\sigma\left(\mathbb{E}_\eta\sup_{\bar{f}\in\mathcal{F}}\left[\frac{1}{n}\sum_{i=1}^n\eta_i\bar{f}(x_i)-C\bar{f}(x_i)^2\right]\right)\\&\le\inf_{\delta\ge0}4\sigma\left\{\frac{(1/2\sigma)\sum_{j=1}^m\log\mathcal{N}(\mathcal{G}_j,L^2(\mathbb{P}_n),\delta/mL)}{n}+\frac{12}{\sqrt{n}}\int_0^\delta\sqrt{\sum_{j=1}^m\log\mathcal{N}\left(\mathcal{G}_j,L^2(\mathbb{P}_n),\frac{\epsilon}{mL}\right)}d\epsilon\right\}.\end{aligned}$$

which is the desired result. □

EC.8. Proof of Theorem 2

Recall that by definition we have $\hat{f}_n\in\arg\min_{f\in\mathcal{F}}\|f-y\|^2_{L^2(\mathbb{P}_n)}$. Notice that $\hat{f}_n$ is actually the projection of $y=\{y_1,\cdots,y_n\}$ onto $\mathcal{F}|_{x_1,\cdots,x_n}$. From the convexity of $\mathcal{F}|_{x_1,\cdots,x_n}$ and Pythagorean theorem, we have

$$\|\hat{f}_n-y\|^2_{L^2(\mathbb{P}_n)}\le\|f-y\|^2_{L^2(\mathbb{P}_n)}-\|\hat{f}_n-f\|^2_{L^2(\mathbb{P}_n)} \tag{EC.2}$$

that holds for any $f \in \mathcal{F}$.

The LHS of Eq. (EC.2) can be further expanded by adding and subtracting f^* as

$$\begin{aligned}\|\hat{f}_n - y\|^2_{L^2(\mathbb{P}_n)} &= \|\hat{f}_n - f^* + f^* - y\|^2_{L^2(\mathbb{P}_n)} \\ &= \|\hat{f}_n - f^*\|_{L^2(\mathbb{P}_n)} + \|f^* - y\|^2_{L^2(\mathbb{P}_n)} + 2\langle \eta, f^* - \hat{f}_n\rangle_{L^2(\mathbb{P}_n)}. \end{aligned} \tag{EC.3}$$

Since the RHS of Eq. (EC.2) allows any f, we choose $\tilde{f} = \arg\min_{f\in\mathcal{F}} \|f - f^*\|^2_{L^2(\mathbb{P}_n)}$ and can get

$$\begin{aligned}\|\tilde{f} - y\|^2_{L^2(\mathbb{P}_n)} - \|\hat{f}_n - \tilde{f}\|^2_{L^2(\mathbb{P}_n)} &= \|\tilde{f} - f^* + f^* - y\|^2_{L^2(\mathbb{P}_n)} - \|\hat{f}_n - \tilde{f}\|^2_{L^2(\mathbb{P}_n)} \\ &= \|\tilde{f} - f^*\|^2_{L^2(\mathbb{P}_n)} + \|f^* - y\|^2_{L^2(\mathbb{P}_n)} + 2\langle \eta, f^* - \tilde{f}\rangle_{L^2(\mathbb{P}_n)} - \|\hat{f}_n - \tilde{f}\|^2_{L^2(\mathbb{P}_n)}. \end{aligned} \tag{EC.4}$$

Put Eqs. (EC.2), (EC.3) and (EC.4) together, and we can have

$$\begin{aligned}\|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P}_n)} - \|\tilde{f} - f^*\|^2_{L^2(\mathbb{P}_n)} &\le 2\langle \eta, \hat{f}_n - \tilde{f}\rangle_{L^2(\mathbb{P}_n)} - \|\hat{f}_n - \tilde{f}\|^2_{L^2(\mathbb{P}_n)} \\ &\le \sup_{\bar{f}\in\mathcal{F}-\tilde{f}^*} \left[\frac{1}{n}\sum_{i=1}^n 2\eta_i \bar{f}(x_i) - \bar{f}(x_i)^2\right].\end{aligned}$$

Now, taking expectation with repect to η on both sides of the inequality, on the LHS we have the metric of interest, and on the RHS we get the offset sub-Gaussian complexity. Using Lemma EC.1, we take $C = 1/2\sigma$ and $\alpha = 0$ and multiplying both sides by 2σ would give us

$$\begin{aligned}\mathbb{E}_\eta\|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P}_n)} - \|\tilde{f} - f^*\|^2_{L^2(\mathbb{P}_n)} &\le 2\sigma\left(\mathbb{E}_\eta \sup_{\bar{f}\in\mathcal{F}}\left[\frac{1}{n}\sum_{i=1}^n \eta_i \bar{f}(x_i) - C\bar{f}(x_i)^2\right]\right) \\ &\le \inf_{\delta\ge 0} 2\sigma\Bigg\{\frac{(1/\sigma)\sum_{j=1}^m \log\mathcal{N}(\mathcal{G}_j, L^2(\mathbb{P}_n), \delta/mL)}{n} \\ &\qquad + \frac{12}{\sqrt{n}}\int_0^\delta \sqrt{\sum_{j=1}^m \log\mathcal{N}\left(\mathcal{G}_j, L^2(\mathbb{P}_n), \frac{\epsilon}{mL}\right)}\, d\epsilon\Bigg\}.\end{aligned}$$

EC.9. Proof of Theorem 3

We start by quoting the bounded difference inequality. We say a function $\phi(x_1, \cdots, x_n) : \prod_{i=1}^n \mathcal{X}_i \to \mathbb{R}$ satisfies the bounded difference property, if substituting the value of the ith variable changes the value of ϕ by at most c_i. Formally speaking, for all $x_i, x_i' \in \mathcal{X}_i$, and for all $x_k \in \mathcal{X}_k, k = 1, \cdots, i-1, i+1, \cdots, n$, we have

$$|\phi(x_1, \cdots, x_{i-1}, x_i, x_{i+1}, \cdots, x_n) - \phi(x_1, \cdots, x_{i-1}, x_i', x_{i+1}, \cdots, x_n)| \le c_i.$$

The bounded difference inequality gives a non-asymptotic bounds on how far function value of ϕ given random input deviates from its expectation.

LEMMA EC.2 **(Wainwright (2019), Corollary 2.21)**. *Consider i.i.d. variables $X_1, \cdots, X_n$. Suppose that function ϕ satisfies the bounded difference property, then with probability $1-\zeta$, we have*

$$\phi(X_1, \cdots, X_n) - \mathbb{E}\phi(X_1, \cdots, X_n) \leq \sqrt{\frac{\sum_{i=1}^n c_i^2}{2} \log \frac{1}{2\zeta}}. \tag{EC.5}$$

We now bridge the gap between the random design and the fixed design as following,

$$\begin{aligned}
&\mathbb{E}_{X,\eta}\|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P}_X)} - \inf_{f\in\mathcal{F}}\|f - f^*\|^2_{L^2(\mathbb{P}_X)} \\
=&\left(\mathbb{E}_{\eta}\|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P}_n)} - \inf_{f\in\mathcal{F}}\|f - f^*\|^2_{L^2(\mathbb{P}_n)}\right) \\
&+\left(\mathbb{E}_{X,\eta}\|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P}_X)} - \inf_{f\in\mathcal{F}}\|f - f^*\|^2_{L^2(\mathbb{P}_X)}\right) - \left(\mathbb{E}_{\eta}\|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P}_n)} - \|\tilde{f}_n - f^*\|^2_{L^2(\mathbb{P}_n)}\right) \\
\leq&\left(\mathbb{E}_{\eta}\|\hat{f}_n - f^*\|^2_{L^2(\mathbb{P}_n)} - \inf_{f\in\mathcal{F}}\|f - f^*\|^2_{L^2(\mathbb{P}_n)}\right) \\
&+\sup_{f'\in\mathcal{F}}\left[\|f' - f^*\|^2_{L^2(\mathbb{P}_X)} - \|f' - f^*\|^2_{L^2(\mathbb{P}_n)} + \inf_{f\in\mathcal{F}}\|f - f^*\|^2_{L^2(\mathbb{P}_n)} - \inf_{f\in\mathcal{F}}\|f - f^*\|^2_{L^2(\mathbb{P}_X)}\right].
\end{aligned} \tag{EC.6}$$

Here we note that all the terms involving the $L^2(\mathbb{P}_n)$ norm are random, because the norm depends on the randomly sampled features $X_1, \cdots, X_n$. The first term is the metric for the mis-specified, fixed design case. Although it is a random variable right now, from Theorem 2 we know that it has a upper bound that holds uniformly across all realizations of features $\{x_1, \cdots, x_n\}$.

Therefore, we only need to elaborate on the second term of Eq. (EC.6). For simplicity, we denote the second term of Eq. (EC.6) as $\phi(X_1, \cdots, X_n)$. Now we show that this term satisfies the bounded difference property. Because we have the uniform boundedness condition $|f| \leq 1$, we have that $(f - f')^2 \leq 4$. Suppose that we change only the ith observation from $x_{i,1}$ to $x_{i,2}$. First we look at $\inf_{f\in\mathcal{F}}\|f - f^*\|^2_{L^2(\mathbb{P}_X)}$. This term stays constant when x_i changes.

Then we look at $\mathbb{E}_{\eta}\|f' - f^*\|^2_{L^2(\mathbb{P}_n)}$. Let $\mathbb{P}_{n,1}, \mathbb{P}_{n,2}$ be the empirical measure of $\{\cdots, x_{i,1}, \cdots\}$ and $\{\cdots, x_{i,2}, \cdots\}$ respectively. The two sets of input features differ only at x_i. Let

$$\begin{aligned}
f_1 &= \operatorname*{arginf}_{f\in\mathcal{F}} \|f - f^*\|^2_{L^2\left(\mathbb{P}_{n,1}\right)}, \\
f_2 &= \operatorname*{arginf}_{f\in\mathcal{F}} \|f - f^*\|^2_{L^2\left(\mathbb{P}_{n,2}\right)}.
\end{aligned}$$

Then we have

$$\begin{aligned}
\|f_2 - f^*\|_{L^2\left(\mathbb{P}_{n,2}\right)} &\leqslant \|f_1 - f^*\|_{L^2\left(\mathbb{P}_{n,2}\right)}, \\
\|f_1 - f^*\|_{L^2\left(\mathbb{P}_{n,1}\right)} &\leqslant \|f_2 - f^*\|_{L^2\left(\mathbb{P}_{n,1}\right)}.
\end{aligned}$$

Notice that

$$\|f_2 - f^*\|_{L^2\left(\mathbb{P}_{n,2}\right)} \leqslant \|f_1 - f^*\|_{L^2\left(\mathbb{P}_{n,2}\right)} \leqslant \|f_1 - f^*\|_{L^2\left(\mathbb{P}_{n,1}\right)} + \frac{4}{n}.$$

Thus

$$\|f_2 - f^*\|_{L^2(\mathbb{P}_n,2)} - \|f_1 - f^*\|_{L^2(\mathbb{P}_n,1)} \leqslant \frac{4}{n}.$$

And from symmetry we have

$$\|f_1 - f^*\|_{L^2(\mathbb{P}_n,1)} - \|f_2 - f^*\|_{L^2(\mathbb{P}_n,2)} \leqslant \frac{4}{n}.$$

Therefore we know that this term changes by no more than $4/n$. A similar argument shows that the first two terms also changes by no more than $4/n$. Combining the results, we know that one observation x_i changes, ϕ can change by no more than $8/n$. Bounded difference inequality implies that with probability at least $1-\zeta$,

$$\phi(X_1, \cdots, X_n) \leq \mathbb{E}[\phi(X_1, \cdots, X_n)] + 4\sqrt{\frac{2\log(1/\zeta)}{n}}. \tag{EC.7}$$

Now we use a technique called symmetrization to further upper bound $\mathbb{E}[\phi(X_1, \cdots, X_n)]$. Before starting the symmetrization argument, it is important to notice that

$$\begin{aligned}
\|f\|_{L^2(\mathbb{P}_X)} &= \mathbb{E}_X\left[f(X)^2\right] \\
&= \frac{1}{n}\sum_{i=1}^n \mathbb{E}_{X_i}\left[f(X_i)^2\right] \\
&= \mathbb{E}_{X_i}\left[\frac{1}{n}\sum_{i=1}^n f(X_i)^2\right] \\
&= \mathbb{E}_X \|f\|_{L^2(\mathbb{P}_n)}.
\end{aligned}$$

In other words, the expectation of the empirical distribution L^2 norm recovers the actual distribution L^2 norm. Since $\mathbb{E}_X[\inf_{f\in\mathcal{F}}\|f-f^*\|^2_{L^2(\mathbb{P}_n)}] \leq \inf_{f\in\mathcal{F}}\mathbb{E}_X[\|f-f^*\|^2_{L^2(\mathbb{P}_n)}] = \inf_{f\in\mathcal{F}}\|f-f^*\|^2_{L^2(\mathbb{P}_n)}$, this third subterm's upper bound cancels out the fourth subterm, and we have

$$\begin{aligned}
\mathbb{E}_X[\phi(X_1, \cdots, X_n)] &\leq \mathbb{E}_X \sup_{f'\in\mathcal{F}}\left[\|f'-f^*\|^2_{L^2(\mathbb{P}_X)} - \|f'-f^*\|^2_{L^2(\mathbb{P}_n)}\right] \\
&= \mathbb{E}_X \sup_{f'\in\mathcal{F}}\left[\mathbb{E}_{X'}\left[\frac{\sum_{i=1}^n (f'(X_i')-f^*(X_i'))^2}{n}\right] - \frac{\sum_{i=1}^n (f'(X_i)-f^*(X_i))^2}{n}\right] \\
&= \mathbb{E}_X \sup_{f'\in\mathcal{F}}\mathbb{E}_{X'}\left[\frac{\sum_{i=1}^n (f'(X_i')-f^*(X_i'))^2}{n} - \frac{\sum_{i=1}^n (f'(X_i)-f^*(X_i))^2}{n}\right] \\
&\leq \mathbb{E}_{X,X'} \sup_{f'\in\mathcal{F}}\left[\frac{\sum_{i=1}^n (f'(X_i')-f^*(X_i'))^2}{n} - \frac{\sum_{i=1}^n (f'(X_i)-f^*(X_i))^2}{n}\right] \\
&= \mathbb{E}_{X,X',\tau} \sup_{f'\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n\left[\tau_i(f'(X_i')-f^*(X_i'))^2 - (f'(X_i)-f^*(X_i))^2)\right] \\
&\leq 2\mathbb{E}_{X,X',\tau} \sup_{f'\in\mathcal{F}}\frac{1}{n}\sum_{i=1}^n\left[\tau_i[(f'(X_i)-f^*(X_i))^2]\right] \\
&= 2\mathbb{E}_{X,X',\tau} \sup_{f'\in\mathcal{F}-f^*}\frac{1}{n}\sum_{i=1}^n\left[\tau_i(f'(X_i))^2\right],
\end{aligned} \tag{EC.8}$$

where $X_1' \cdots, X_n'$ are independent random variables with the same distribution as $X_1, \cdots, X_n$, and τ_is are independent Rademacher random variables (variables with $1/2$ chance being $+1$, and $1/2$

chance being -1). The second inequality holds due to the convexity of the supreme and the tower property. The third equality and the third inequality are because of the symmetric structure of X_i' and X_i. Eq. (EC.8) is the Rademacher complexity of the function class $(\mathcal{F}-f^*)^2 := \{f'^2 : f' \in \mathcal{F}-f^*\}$ under $X_1, \cdots, X_n$, which can be finally reduced to the function class $\mathcal{F}-f^*$ as following. Conditional on $\tau_1, \cdots, \tau_{n-1}$,

$$
\begin{aligned}
&\mathbb{E}_{\tau_n} \sup_{f'\in\mathcal{F}-f^*} \sum_{i=1}^{n} \left[\tau_i (f'(X_i))^2\right] \\
&= \frac{1}{2}\left(\sup_{f'\in\mathcal{F}-f^*} \left(\sum_{i=1}^{n-1} \left[\tau_i (f'(X_i))^2\right] + (f'(X_n))^2 \right) + \sup_{f''\in\mathcal{F}-f^*} \left(\sum_{i=1}^{n-1} \left[\tau_i (f''(X_i))^2\right] - (f''(X_n))^2 \right) \right) \\
&\le \frac{1}{2} \sup_{f',f''\in\mathcal{F}-f^*} \left(\sum_{i=1}^{n-1} \left[\tau_i (f'(X_i))^2\right] + \sum_{i=1}^{n-1} \left[\tau_i (f''(X_i))^2\right] + 2|(f'(X_n)) - (f''(X_n))| \right) \\
&= \frac{1}{2} \sup_{f',f''\in\mathcal{F}-f^*} \left(\sum_{i=1}^{n-1} \left[\tau_i (f'(X_i))^2\right] + \sum_{i=1}^{n-1} \left[\tau_i (f''(X_i))^2\right] + 2f'(X_n) - 2f''(X_n) \right) \\
&= \frac{1}{2}\left(\sup_{f'\in\mathcal{F}-f^*} \left(\sum_{i=1}^{n-1} \left[\tau_i (f'(X_i))^2\right] + 2f'(X_n) \right) + \sup_{f''\in\mathcal{F}-f^*} \left(\sum_{i=1}^{n-1} \left[\tau_i (f''(X_i))^2\right] - 2f''(X_n) \right) \right) \\
&= \mathbb{E}_{\tau_n} \sup_{f'\in\mathcal{F}-f^*} \sum_{i=1}^{n-1} \left[\tau_i (f'(X_i))^2\right] + 2\tau_n f'(X_n),
\end{aligned}
$$

where the first inequality holds because the $(\cdot)^2$ is 2-Lipschitz on $[-1,1]$, the second equality is based on the symmetric structure of f' and f''. By proceeding, we can have $\mathbb{E}_\tau \sup_{f'\in\mathcal{F}-f^*} \sum_{i=1}^n [\tau_i (f'(X_i))^2] \le 2\mathbb{E}_\tau \sup_{f'\in\mathcal{F}-f^*} \sum_{i=1}^n [\tau_i f'(X_i)]$. Therefore, we can have

$$
\begin{aligned}
2\mathbb{E}_\tau \sup_{f'\in\mathcal{F}-f^*} \frac{1}{n}\sum_{i=1}^{n} \left[\tau_i (f'(X_i))^2\right] &\le 4\mathbb{E}_\tau \sup_{f'\in\mathcal{F}-f^*} \frac{1}{n}\sum_{i=1}^{n} [\tau_i f'(X_i)] \\
&\le \frac{48}{\sqrt{n}} \int_0^1 \sqrt{\log \mathcal{N}(\mathcal{F}, L^2(\mathbb{P}_n), \epsilon)}\, d\epsilon. \qquad \text{(EC.9)}
\end{aligned}
$$

The second inequality is the famous Dudley's integral (Rakhlin (2022), Corollary 2). Putting Eqs. (EC.6), (EC.7), (EC.9) and Proposition 1 together, we finish the proof. □

EC.10. Proof on the L-smoothness of Additive and Multiplicative Model in Example 1.

Argument: (i) the ERM of the additive model $\hat{\mathcal{L}}(\mathbf{g}_1, \cdots, \mathbf{g}_m) = \sum_{i=1}^n (y_i - \sum_{j=1}^m g_{j,i})^2$ is L-smooth with $L = 2m$; (ii) the ERM of the multiplicative model $\hat{\mathcal{L}}(\mathbf{g}_1, \cdots, \mathbf{g}_m) = \sum_{i=1}^n (y_i - \prod_{j=1}^m g_{j,i})^2$ is L-smooth given that every $\mathcal{G}_j$ is uniformly bounded and the range of a function is strictly positive.

Denote $\mathbf{g} = (\mathbf{g}_j)_{j=1}^m$, we have

(i) We have:

$$\begin{aligned}
\hat{\mathcal{L}}\left(\mathbf{g}'\right) &= \sum_{i=1}^{n}\left(y_i - \sum_{j=1}^{m} g'_{j,i}\right)^2 \\
&= \sum_{i=1}^{n}\left(y_i - \sum_{j=1}^{m} g_{j,i} + \sum_{j=1}^{m} g_{j,i} - \sum_{j=1}^{m} g'_{j,i}\right)^2 \\
&= \sum_{i=1}^{n}\left(y_i - \sum_{j=1}^{m} g_{j,i}\right)^2 + 2\sum_{i=1}^{n}\left(y_i - \sum_{j=1}^{m} g_{j,i}\right)\left(\sum_{j=1}^{m} g_{j,i} - \sum_{j=1}^{m} g'_{j,i}\right) + \sum_{i=1}^{n}\left(\sum_{j=1}^{m}\left(g_{j,i} - g'_{j,i}\right)\right)^2 \\
&= \hat{\mathcal{L}}(\mathbf{g}) + \left\langle \nabla\hat{\mathcal{L}}(\mathbf{g}), \mathbf{g}' - \mathbf{g}\right\rangle + \sum_{i=1}^{n}\left(\sum_{j=1}^{m}\left(g_{j,i} - g'_{j,i}\right)\right)^2 \\
&\leqslant \hat{\mathcal{L}}(\mathbf{g}) + \left\langle \nabla\hat{\mathcal{L}}(\mathbf{g}), \mathbf{g}' - \mathbf{g}\right\rangle + \sum_{i=1}^{n}\left[\sum_{j=1}^{m} 1^2 \cdot \sum_{j=1}^{m}\left(g_{j,i} - g'_{j,i}\right)^2\right] \\
&= \hat{\mathcal{L}}(\mathbf{g}) + \left\langle \nabla\hat{\mathcal{L}}(\mathbf{g}), \mathbf{g}' - \mathbf{g}\right\rangle + m\left\|\mathbf{g}' - \mathbf{g}\right\|^2,
\end{aligned}$$

where the inequality is by Cauchy-Schwarz inequality applied to each i.

(ii) We prove the L-smoothness of the Multiplicative model by showing the Hessian $\|\nabla^2\Phi\| \leq L$ on a compact subset of the positive cone. Here, the matrix norm returns the largest singular value of the matrix. First, notice that $\hat{\mathcal{L}}(\mathbf{g})$ is twice differentiable. Given that $\mathcal{G}_j$ is uniformly bounded, we know that $\mathbf{g}_j$ must be bounded. Because of the compactness assumption and $\hat{\mathcal{L}}$ being smooth, it would show that the second order partial derivative is also bounded. Then utilizing Gershgorin circle theorem, we know that the Hessian must have bounded maximal singular value, which leads to Φ being L-smooth. □

EC.11. Proof of Theorem 4: the smooth and convex case

Because the equivalence between Algorithm 1 and Algorithm 2, it suffices to analyze the computational performance of Algorithm 2 when the $\hat{\mathcal{L}}$ is smooth and convex. To avoid confusion between the empirical loss function $\hat{\mathcal{L}}$ and the Lipschitz constant L, we use another notation, Φ, to denote the empirical loss function. This notation is consistent with the notation used in Definition 2. Meanwhile, to make this proof concise, we also adopt the following short-handed notations:

$$\begin{aligned}
&v^t = (\hat{\mathbf{g}}_1^t, \hat{\mathbf{g}}_2^t, \cdots, \hat{\mathbf{g}}_m^t);\\
&w_0^t = v^{t-1} = (\hat{\mathbf{g}}_1^{t-1}, \hat{\mathbf{g}}_2^{t-1}, \cdots, \hat{\mathbf{g}}_m^{t-1});\\
&w_1^t = (\hat{\mathbf{g}}_1^t, \hat{\mathbf{g}}_2^{t-1}, \cdots, \hat{\mathbf{g}}_m^{t-1});\\
&\cdots\\
&w_j^t = (\hat{\mathbf{g}}_1^t, \cdots, \hat{\mathbf{g}}_j^t, \hat{\mathbf{g}}_{j+1}^{t-1}, \cdots, \hat{\mathbf{g}}_m^{t-1});\\
&\cdots\\
&w_m^t = v^t = (\hat{\mathbf{g}}_1^t, \hat{\mathbf{g}}_2^t, \cdots, \hat{\mathbf{g}}_m^t).
\end{aligned}$$

Also, we denote the minimum point(s) of Φ by v^*.

We start by revealing the fact that for a smooth and convex function, its blockwise minimum point is equivalent to global minimum point.

PROPOSITION EC.1 **(Blockwise ≡ global)**. *For a smooth convex function Φ defined on $\prod_{j=1}^{m} \mathcal{G}_j \mid_{x_1,\cdots,x_n}$, a blockwise minimum point is a global minimum point.*

Proof. Suppose that $\tilde{\mathbf{g}} = (\tilde{\mathbf{g}}_1, \cdots, \tilde{\mathbf{g}}_m)$ is a blockwise minimum point, meaning that:

$$\Phi(\tilde{\mathbf{g}}) \leq \Phi(\mathbf{g}_j, \tilde{\mathbf{g}}_{-j}) \quad \forall j, \forall \mathbf{g}_j \in \mathcal{G}_j \mid_{x_1,\cdots,x_n} . \tag{EC.10}$$

Define $\Phi'(v; d)$ as the function's directional derivative at v with direction d. According to the first-order condition, we have:

$$\Phi'(\tilde{\mathbf{g}}; (0, \cdots, \mathbf{g}_j - \tilde{\mathbf{g}}_j, \cdots, 0)) \geq 0 \quad \forall j, \forall \mathbf{g}_i \in \mathcal{G}_j \mid_{x_1,\cdots,x_n} . \tag{EC.11}$$

Note that since $\mathcal{F}$ is a convex set, the directional derivatives are well defined.

We now prove by contradiction. Suppose there exists some $\mathbf{g} = (\mathbf{g}_1, \cdots, \mathbf{g}_m)$ such that $\Phi(\mathbf{g}) < \Phi(\tilde{\mathbf{g}})$. In that case, we will have $\Phi'(\tilde{\mathbf{g}}; \mathbf{g} - \tilde{\mathbf{g}}) < 0$, because according to the convexity of Φ, for $\varepsilon \in (0, 1)$ we have:

$$\frac{\Phi(\tilde{\mathbf{g}} + \varepsilon(\mathbf{g} - \tilde{\mathbf{g}})) - \Phi(\tilde{\mathbf{g}})}{\varepsilon} \leq \frac{\Phi(\tilde{\mathbf{g}} + (\mathbf{g} - \tilde{\mathbf{g}})) - \Phi(\tilde{\mathbf{g}})}{1} < 0.$$

But from Eq. (EC.11), we have

$$\Phi'(\tilde{\mathbf{g}}; \mathbf{g} - \tilde{\mathbf{g}}) = \sum_{j=1}^{m} \Phi'(\tilde{\mathbf{g}}; (0, \cdots, \mathbf{g}_j - \tilde{\mathbf{g}}_j, \cdots, 0)) \geq 0,$$

which leads to a contradiction. □

Because $\{\Phi(v^t)\}$ is lower bounded and monotonically decreases, it must have a limit. Similarly, because the level set S is closed and bounded, we know that $\{v^t\}$ must have a limit point. The remaining step is to prove that any limit point is a blockwise minimum point, where we refer to Theorem 4.1 in Tseng (2001).

PROPOSITION EC.2 **(Tseng (2001), Theorem 4.1, adapted version)**. *Assume that the level set S is compact and that Φ is continuous on S. Then, the sequence $\{w_0^1, w_1^1, \cdots, w_m^1, w_1^2, \cdots, w_m^2, \cdots, w_1^T, \cdots, w_m^T\}$ generated by Algorithm 2 is defined and bounded. If $\Phi(\mathbf{g}_1, \cdots, \mathbf{g}_m)$ is also smooth and convex in S, then every limit point of the sequence is a blockwise minimum point of Φ.*

Then, it follows from Proposition EC.1 that any limit point that comes from the sequence is also the global minimum point, and therefore the corresponding limit point of submodel sequences generated by Algorithm 1 must also be the global minimum point of $\hat{\mathcal{L}}(g_1, \cdots, g_m)$.

Also, notice that because at least a limit point exists, it follows that the limit of the loss sequence $\{\hat{\mathcal{L}}(g_1, \cdots, g_m)\}$ must converge to the global minimum $\{\hat{\mathcal{L}}(\hat{g}_1, \cdots, \hat{g}_m)\}$. □

EC.12. Proof of Theorem 4: the L-smooth and convex case

We adopt the same notation convention as Section EC.11.

Because the equivalence between Algorithm 1 and Algorithm 2, it suffices to analyze the computational performance of Algorithm 2 when the $\hat{L}$ is convex and Lipschitz smooth. To avoid confusion between the empirical loss function $\hat{L}$ and the Lipschitz constant L, we use another notation, Φ, to denote the empirical loss function. This notation is consistent with the notation used in Definition 2. Meanwhile, to make this proof concise, we also adopt the following short-handed notations:

$$
\begin{aligned}
&v^t = (\hat{\mathbf{g}}_1^t, \hat{\mathbf{g}}_2^t, \cdots, \hat{\mathbf{g}}_m^t);\\
&w_0^t = v^{t-1} = (\hat{\mathbf{g}}_1^{t-1}, \hat{\mathbf{g}}_2^{t-1}, \cdots, \hat{\mathbf{g}}_m^{t-1});\\
&w_1^t = (\hat{\mathbf{g}}_1^t, \hat{\mathbf{g}}_2^{t-1}, \cdots, \hat{\mathbf{g}}_m^{t-1});\\
&\cdots\\
&w_j^t = (\hat{\mathbf{g}}_1^t, \cdots, \hat{\mathbf{g}}_j^t, \hat{\mathbf{g}}_{j+1}^{t-1}, \cdots, \hat{\mathbf{g}}_m^{t-1});\\
&\cdots\\
&w_m^t = v^t = (\hat{\mathbf{g}}_1^t, \hat{\mathbf{g}}_2^t, \cdots, \hat{\mathbf{g}}_m^t).
\end{aligned}
$$

Also, we denote the minimum point(s) of Φ by v^*.

The proof repeatedly use the following fact:

$$
\begin{aligned}
\text{if} \quad & \hat{\mathbf{g}}_j^t = \underset{\gamma_j \in \mathcal{G}_j|_{x_1,\cdots,x_n}}{\arg\min} \Phi(\hat{\mathbf{g}}_1^t, \cdots, \hat{\mathbf{g}}_{j-1}^t, \gamma_j, \hat{\mathbf{g}}_{j+1}^{t-1}, \cdots, \hat{\mathbf{g}}_m^{t-1}),\\
\text{then} \quad & \langle \nabla_j \Phi(w_j^t), \gamma_j - \hat{\mathbf{g}}_j^t \rangle \geq 0, \quad \forall \gamma_j \in \mathcal{G}_j\,|_{x_1,\cdots,x_n}.
\end{aligned}
\tag{EC.12}
$$

and vice versa. This fact is easily verified by examining the first-order Taylor expansion at w_j^t, and we call it the *optimality condition*.

Our proof consists of three main steps, which is similar to the typical proof strategy employed in the computational guarantee theorems for gradient descent algorithms:

1. Successive descent: estimate the descent of the objective after each iteration;

2. Gap towards the optimal value: estimate the cost yet to be minimized (cost-to-go) after each iteration;

3. Convergence rate: combine the above two estimates to obtain the boundary of cost-to-go.

Successive descent: We first introduce a condition equivalent to the L-smooth condition:

$$
\Phi(v^1) \geq \Phi(v^2) + \langle \nabla\Phi(v^2), v^1 - v^2 \rangle + \frac{1}{2L}\|\nabla\Phi(v^2) - \nabla\Phi(v^1)\|^2.
$$

Therefore, utilizing this inequality, and combining with the optimality condition, for $i = 1, \cdots, n$,

$$
\Phi(w_{j-1}^t) - \Phi(w_j^t) \geq \langle \nabla_j \Phi(w_j^t), \hat{\mathbf{g}}_j^{t-1} - \hat{\mathbf{g}}_j^t \rangle + \frac{1}{2L}\|\nabla\Phi(w_{j-1}^t) - \nabla\Phi(w_j^t)\|^2.
$$

Adding all the inequalities together, we have:

$$\Phi(v^{t-1}) - \Phi(v^t) \geq \frac{1}{2L} \sum_{j=1}^{m} \|\nabla\Phi(w_{j-1}^t) - \nabla\Phi(w_j^t)\|^2. \tag{EC.13}$$

Gap towards the optimal value: From the convexity of Φ, combining with the optimality condition, we have:

$$\begin{aligned}
\Phi(v^*) - \Phi(v^t) &\geq \sum_{j=1}^{m} \langle \nabla_j \Phi(v^t), \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \rangle \\
&= \sum_{j=1}^{m} \langle \nabla_j \Phi(v^t) - \nabla_j \Phi(w_j^t), \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \rangle + \sum_{i=1}^{n} \langle \nabla_j \Phi(w_j^t), \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \rangle \\
&\geq \sum_{j=1}^{m} \langle \nabla_j \Phi(v^t) - \nabla_j \Phi(w_j^t), \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \rangle \\
&\geq -\sum_{j=1}^{m} \|\nabla_j \Phi(v^t) - \nabla_j \Phi(w_j^t)\| \cdot \|\hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t\|.
\end{aligned}$$

Therefore, we have:

$$\begin{aligned}
\Phi(v^t) - \Phi(v^*) &\leq \sum_{j=1}^{m} \|\nabla_j \Phi(v^t) - \nabla_j \Phi(w_j^t)\| \cdot \|\hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t\| \\
&= \sum_{j=1}^{m} \|\nabla_j \Phi(v^t) - \nabla_j \Phi(w_{n-1}^t) + \nabla_j \Phi(w_{n-1}^t) - \cdots - \nabla_j \Phi(w_j^t)\| \cdot \|\hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t\| \\
&\leq \sum_{j=1}^{m} \sum_{k=j+1}^{m} \|\nabla_j \Phi(w_{k-1}^t) - \nabla_j \Phi(w_k^t)\| \cdot \|\hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t\| \\
&\leq \sum_{j=1}^{m} \sum_{k=1}^{m} \|\nabla_j \Phi(w_{k-1}^t) - \nabla_j \Phi(w_k^t)\| \cdot \|\hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t\| \\
&\leq R \sum_{j=1}^{m} \sum_{k=1}^{m} \|\nabla_j \Phi(w_{k-1}^t) - \nabla_j \Phi(w_k^t)\|.
\end{aligned}$$

Define $\varepsilon_t = \Phi(v^t) - \Phi(v^*)$, squaring both sides, and from Cauchy-Schwarz inequality $(a_1 + a_2 + \cdots + a_n)^2 \leq n(a_1^2 + a_2^2 + \cdots + a_n^2)$, we have the following:

$$\varepsilon_t^2 \leq R^2 n^2 \sum_{j=1}^{n} \sum_{i=1}^{n} \|\nabla_i \Phi(w_{j-1}^t) - \nabla_i \Phi(w_j^t)\|^2 = R^2 n^2 \sum_{j=1}^{n} \|\nabla\Phi(w_{j-1}^t) - \nabla\Phi(w_j^t)\|^2. \tag{EC.14}$$

Convergence rate: Define $\sigma = \frac{1}{2LR^2m^2}$, combining Eq. (EC.13) and Eq. (EC.14), we have:

$$\varepsilon_{t-1} - \varepsilon_t \geq \sigma \varepsilon_t^2.$$

and therefore:

$$\varepsilon_t \leq \frac{-1 + \sqrt{1 + 4\sigma\varepsilon_{t-1}}}{2\sigma}.$$

We now prove the sublinear convergence rate by induction. Define $c = \max\{4\sigma - 2, \varepsilon_0, 2\}$, then for ε_1, we have:

$$\varepsilon_1 \le \frac{-1+\sqrt{1+4\sigma\varepsilon_0}}{2\sigma} \le \frac{-1+\sqrt{1+4\sigma c}}{2\sigma} = \frac{2c}{1+\sqrt{1+4\sigma c}} \le \frac{2c}{1+|4\sigma-1|},$$

in which the last inequality comes from the fact that $c \ge 4\sigma - 1$, then $\sqrt{1+4\sigma c} \ge \sqrt{1-4\sigma+16\sigma^2} \ge \sqrt{(1-4\sigma)^2}$.

For the above equation, if $4\sigma - 1 \ge 0$, then naturally $\varepsilon_2 \le \frac{c}{2\sigma}$. If $4\sigma - 1 < 0$, then $2 > 8\sigma$, therefore $\varepsilon_2 \le \frac{2c}{2-4\sigma} \le \frac{2c}{8\sigma-4\sigma} = \frac{c}{2\sigma}$.

Next, if we assume that $\varepsilon_{t-1} \le c/\sigma t$, then

$$\varepsilon_{t+1} \le \frac{-1+\sqrt{1+4\sigma\cdot\frac{c}{\sigma t}}}{2\sigma} = \frac{2c}{t\sigma(1+\sqrt{1+\frac{4c}{t}})} \le \frac{2c}{\sigma(t+\sqrt{t^2+4ct})}.$$

Because $c \ge 2$ and $t \ge 1$, we have $4ct \ge 4t+4$, which leads to:

$$\varepsilon_{t+1} \le \frac{2c}{\sigma(t+\sqrt{t^2+4t+4})} \le \frac{c}{\sigma(t+1)}.$$

□

EC.13. Proof of Theorem 4: the L-smooth and μ-strongly convex case

The proof strategy is the same as the L-smooth and convex case, consisting of the same three steps. However, the specific term used to connect the successive descent term and the gap to the optimal value term are different. We adopt the same notation convention as Section EC.11.

Successive descent: from the μ-strongly convex condition, for each $i = 1, 2, \cdots, n$, we have

$$\Phi(w_{j-1}^t) - \Phi(w_j^t) \ge \langle \nabla_i \Phi(w_j^t), \hat{\mathbf{g}}_j^{t-1} - \hat{\mathbf{g}}_j^t \rangle + \frac{\mu}{2}\|\hat{\mathbf{g}}_j^{t-1} - \hat{\mathbf{g}}_j^t\|^2.$$

Adding all the inequalities together,

$$\Phi(v^{t-1}) - \Phi(v^t) \ge \frac{\mu}{2}\sum_{j=1}^m \|\hat{\mathbf{g}}_j^{t-1} - \hat{\mathbf{g}}_j^t\|^2. \tag{EC.15}$$

Note that Eq.(EC.15) holds because the inner product term are non-negative according to the optimality condition.

Gap towards the optimal value: From μ-strongly convexity, combining with the optimality condition,

$$\begin{aligned}
&\Phi(v^*) - \Phi(v^t) \\
&\geqslant \sum_{j=1}^m \left\langle \nabla_j \Phi(v^t), \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \right\rangle + \frac{\mu}{2}\sum_{j=1}^m \left\|\hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t\right\|^2
\end{aligned}$$

$$
\begin{aligned}
&= \sum_{j=1}^{m} \left\langle \nabla_j \Phi\left(v^t\right) - \nabla_j \Phi\left(w_j^t\right), \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \right\rangle + \sum_{j=1}^{m} \left\langle \nabla_j \Phi\left(w_j^t\right), \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \right\rangle + \frac{\mu}{2} \sum_{j=1}^{m} \left\| \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \right\|^2 \\
&\geqslant \sum_{j=1}^{m} \left\langle \nabla_j \Phi\left(v^t\right) - \nabla_j \Phi\left(w_j^t\right), \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \right\rangle + \frac{\mu}{2} \sum_{j=1}^{m} \left\| \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \right\|^2 \\
&\geqslant \sum_{j=1}^{m} - \left\| \nabla_j \Phi\left(v^t\right) - \nabla_j \Phi\left(w_j^t\right) \right\| \cdot \left\| \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \right\| + \frac{\mu}{2} \left\| \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \right\|^2 \\
&\geqslant \sum_{j=1}^{m} -L \left\| v^t - w_j^t \right\| \cdot \left\| \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \right\| + \frac{\mu}{2} \left\| \hat{\mathbf{g}}_j^* - \hat{\mathbf{g}}_j^t \right\|^2 \\
&\geqslant \sum_{j=1}^{m} \min_{\alpha_j} -L \left\| v^t - w_j^t \right\| \alpha_j + \frac{\mu}{2} \alpha_j^2 \\
&\geqslant \sum_{j=1}^{m} -\frac{L^2}{2\mu} \left\| v^t - w_j^t \right\|^2 \\
&\geqslant -\frac{L^2}{2\mu} \sum_{j=1}^{m} \sum_{k=j+1}^{m} \left\| \hat{\mathbf{g}}_k^t - \hat{\mathbf{g}}_k^{t-1} \right\|^2 \\
&\geqslant -\frac{L^2}{2\mu} \sum_{j=1}^{m} \sum_{k=1}^{m} \left\| \hat{\mathbf{g}}_k^t - \hat{\mathbf{g}}_k^{t-1} \right\|^2 \\
&\geqslant -\frac{mL^2}{2\mu} \sum_{k=1}^{m} \left\| \hat{\mathbf{g}}_k^t - \hat{\mathbf{g}}_k^{t-1} \right\|^2 .
\end{aligned}
$$

Therefore, we have:

$$\Phi(v^t) - \Phi(v^*) \leq \frac{mL^2}{2\mu} \sum_{j=1}^{m} \|\hat{\mathbf{g}}_j^t - \hat{\mathbf{g}}_j^{t-1}\|^2. \tag{EC.16}$$

Convergence rate: define $\varepsilon_t = \Phi(v^t) - \Phi(v^*)$, combining Eq. (EC.15) and Eq. (EC.16), we have:

$$\varepsilon_{t-1} - \varepsilon_t \geq \frac{\mu}{2} \|v^t - v^{t-1}\|^2 \geq \frac{\mu^2}{mL^2} \varepsilon_t,$$

which leads to:

$$\varepsilon_t \leq \left(\frac{mL^2}{mL^2 + \mu^2} \right) \varepsilon_{t-1} \leq \left(\frac{mL^2}{mL^2 + \mu^2} \right)^t \varepsilon_0.$$

□

EC.14. Proof of Theorem 5: the smooth and convex case

Similar to the exact oracle scenario, it suffices to analyze the computational performance of Algorithm 2, which is essentially a block coordinate descent algorithm. We use the same short-handed notation as EC.11.

Because we assume level set S is compact, and the inexact oracle must reduce the objective in each iteration, we know that the $\{v^t\}$ sequence stays in S, and therefore limit points exist within the sequence. Let us denote a limit point of $\{v^t\}$ by z.

We now show that z must be a blockwise minimum point. If we can do that, from Proposition EC.1 we know it must be a global minimum point and the theorem is proved.

Consider the subsequence $\{v^r\}_{r\in R}$ that converges to z. Here $R\subseteq\mathcal{N}$. Recall that we use w_j^t to denote the vector after the update of jth block at iteration t. By further passing to subsequences if necessary, we would have $\{w_j^r\}_{r\in R}$ converge to some z_j, for all $j=1,\cdots,m$. From the guarantee of the inexact oracle (Assumption 2), we have that

$$\begin{aligned}\Phi(w_j^r) &\le \min_{\gamma_j\in\mathcal{G}_j|_{x_1,\cdots,x_n}} \Phi(\mathbf{g}_1^r,\cdots,\mathbf{g}_{j-1}^r,\gamma_j,\mathbf{g}_{j+1}^{r-1},\cdots,\mathbf{g}_m^{r-1}) + \max_{v\in S}|\nabla_j\Phi(v)|\,\varepsilon_r\\ &\le \Phi(\mathbf{g}_1^r,\cdots,\mathbf{g}_{j-1}^r,\gamma_j,\mathbf{g}_{j+1}^{r-1},\cdots,\mathbf{g}_m^{r-1}) + \max_{v\in S}|\nabla_j\Phi(v)|\,\varepsilon_r\end{aligned}$$

for any $\gamma_j\in\mathcal{G}_j|_{x_1,\cdots,x_n}$. Here, $\max_{v\in S}|\nabla_j\Phi(v)|$ exists because S is compact. Taking limits on both sides of this inequality would lead to $\Phi(z_j)$ being a minimum point on only block j. It turns out this intermediate result is enough to show that $z=z_m$ is a blockwise minimum point on all blocks. The full proof is described in detail in Tseng (2001), p.482. Here we give a sketch of how the remaining proofs work.

Now we consider the properties of $z_1,\cdots,z_m$. A simple sandwich argument would show that $\Phi(z_1)=\Phi(z_2)=\cdots=\Phi(z_m)$. For any $j=2,\cdots,m$, we can see that z_j and z_{j-1} differs only on block j. Adding the fact that $\Phi(z_{j-1})=\Phi(z_j)$, we would have that z_{j-1} is also a minimum point on block j. The final step is to use an induction argument to show that for any $j=1,\cdots,m$, z_j is not only optimal on block j, but also optimal on all previous blocks $k=1,\cdots,j$. Taking the fact $z=z_m$, we finally proved that z is a blockwise minimum point. □

EC.15. Proof of Theorem 5: the L-smooth and μ-strongly convex case

Similar to the exact oracle scenario, it suffices to analyze the computational performance of Algorithm 2, which is essentially a block coordinate descent algorithm.

We start with a lemma that shows the sensitivity of minimizers at a given block with respect to perturbations at other blocks.

Lemma EC.3. *Assume that function $\Phi(x,y)$ that is L-smooth and μ-strongly convex. Consider the constrained minimization problem at y_1,y_2. Denote by $x_1=\arg\min_{x\in X}\Phi(x,y_1)$ and $x_2=\arg\min_{x\in X}\Phi(x,y_2)$. If $\|y_1-y_2\|\le\delta$, then we have $\|x_1-x_2\|\le L\delta/\mu$.*

Proof. For constrained optimization, the first-order optimality condition is given by:

$$\langle\nabla_x\Phi(x_i,y_i),\,x-x_i\rangle\ge 0,\qquad\forall x\in X, i=1,2.$$

Using this condition, we would have

$$\langle\nabla_x\Phi(x_1,y_1)-\nabla_x\Phi(x_2,y_2),\,x_1-x_2\rangle\le 0.$$

Now we decompose the gradient difference:

$$\nabla_x \Phi(x_1, y_1) - \nabla_x \Phi(x_2, y_2) = \underbrace{\nabla_x \Phi(x_1, y_1) - \nabla_x \Phi(x_1, y_2)}_{\text{(A)}} + \underbrace{\nabla_x \Phi(x_1, y_2) - \nabla_x \Phi(x_2, y_2)}_{\text{(B)}}$$

Consider the term (A), we have:

$$\langle (\text{A}), x_2 - x_1 \rangle \leq \|(\text{A})\| \|x_1 - x_2\| \leq L\delta \|x_1 - x_2\|.$$

because of Cauchy inequality and the Lipschitz smoothness. For term (B), we have strong convexity, which implies strong monotonicity:

$$\langle (\text{B}), x_1 - x_2 \rangle \geq \mu \|x_1 - x_2\|^2.$$

Putting everything together, we would have

$$\mu \|x_1 - x_2\|^2 \leq \langle (\text{B}), x_1 - x_2 \rangle \leq \langle (\text{A}), x_1 - x_2 \rangle \leq L\delta \|x_1 - x_2\|.$$

Assuming $\|x_1 - x_2\| > 0$, divide both sides would give us:

$$\|x_1 - x_2\| \leq \left(\frac{L}{\mu}\right) \delta.$$

□

We now consider the trajectory of intermediate solutions between an inexact oracle and an exact one at iteration t, starting at the same initial point $\mathbf{g}^{t-1}$. We adopt the following notation: denote the exact solution produced by exact oracles at block j by $\mathbf{g}_j^t$, and approximate solutions produced by inexact oracles by $\tilde{\mathbf{g}}_j^t$. Recall that the error margin (error upper bound) at each block is ε_t.

Suppose that we are currently updating the jth block. The previous $j-1$ blocks update has resulted in a difference of

$$\left\|(\tilde{\mathbf{g}}_1^t, \cdots, \tilde{\mathbf{g}}_{j-1}^t, \mathbf{g}_j^{t-1}, \cdots, \mathbf{g}_m^{t-1}) - (\mathbf{g}_1^t, \cdots, \mathbf{g}_{j-1}^t, \mathbf{g}_j^{t-1}, \cdots, \mathbf{g}_m^{t-1})\right\| \leq D_{j-1}.$$

Then, according to Lemma EC.3, we would have

$$\begin{aligned}
\|\tilde{\mathbf{g}}_j^t - \mathbf{g}_j^t\| \leq & \left\| \tilde{\mathbf{g}}_j^t - \underset{\gamma_j \in \mathcal{G}_j | x_1, \cdots, x_n}{\arg\min} \Phi(\tilde{\mathbf{g}}_1^t, \cdots, \tilde{\mathbf{g}}_{j-1}^t, \gamma_j, \mathbf{g}_{j+1}^t, \cdots, \mathbf{g}_m^t) \right\| \\
& + \left\| \underset{\gamma_j \in \mathcal{G}_j | x_1, \cdots, x_n}{\arg\min} \Phi(\tilde{\mathbf{g}}_1^t, \cdots, \tilde{\mathbf{g}}_{j-1}^t, \gamma_j, \mathbf{g}_{j+1}^t, \cdots, \mathbf{g}_m^t) - \mathbf{g}_j^t \right\| \\
\leq & \varepsilon_t + \frac{L}{\mu} D_{j-1}.
\end{aligned}$$

It is worth noting that $\mathbf{g}_j^t \neq \arg\min_{\gamma_j \in \mathcal{G}_j|_{x_1,\cdots,x_n}} \Phi(\tilde{\mathbf{g}}_1^t, \cdots, \tilde{\mathbf{g}}_{j-1}^t, \gamma_j, \mathbf{g}_{j+1}^t, \cdots, \mathbf{g}_m^t)$, because the argmin term is the case of using inexact oracles up to block $j-1$ and only using the exact oracle at block j, while $\mathbf{g}_j^t$ is the result of using exact oracles at all blocks. From the above result we have

$$\begin{aligned} D_j &\leq \left(\varepsilon_t + \frac{L}{\mu} D_{j-1}\right) + D_{j-1} \\ &= \varepsilon_t + \left(1 + \frac{L}{\mu}\right) D_{j-1}. \end{aligned}$$

Therefore, we know that

$$\begin{aligned} D_m &\leq \varepsilon_t + \left(1 + \frac{L}{\mu}\right) D_{m-1} \\ &\leq \varepsilon_t + \left(1 + \frac{L}{\mu}\right) \left[\varepsilon_t + \left(1 + \frac{L}{\mu}\right) D_{m-2}\right] \\ &\cdots \\ &\leq \sum_{k=1}^{m} \left(1 + \frac{L}{\mu}\right)^{m-k} \varepsilon_t \\ &= \frac{(\mu + L)^m - \mu^m}{\mu^{m-1} L} \varepsilon_t. \end{aligned}$$

From this relation, we have that

$$\Phi(\tilde{\mathbf{g}}_1^t, \cdots, \tilde{\mathbf{g}}_m^t) - \Phi(\mathbf{g}_1^t, \cdots, \mathbf{g}_m^t) \leq \max_{\mathbf{g} \in S} \|\nabla \Phi(\mathbf{g})\| D_m = c\varepsilon_t.$$

Now, from Theorem 4, we know that for exact oracles, we have

$$\Phi(\mathbf{g}^t) - \Phi(\mathbf{g}^*) \leq \left(\frac{mL^2}{mL^2 + \mu^2}\right) \left[\Phi(\mathbf{g}^{t-1}) - \Phi(\mathbf{g}^*)\right]$$

Therefore we know

$$\Phi(\tilde{\mathbf{g}}^t) - \Phi(\mathbf{g}^*) \leq \left(\frac{mL^2}{mL^2 + \mu^2}\right) \left[\Phi(\mathbf{g}^{t-1}) - \Phi(\mathbf{g}^*)\right] + c\varepsilon_t,$$

which holds for any starting point $\mathbf{g}^{t-1}$ and any t. Iteratively applying this result would give us the final bound:

$$\begin{aligned} \Phi(\tilde{\mathbf{g}}^t) - \Phi(\mathbf{g}^*) &\leq \left(\frac{mL^2}{mL^2 + \mu^2}\right) \left[\Phi(\mathbf{g}^{t-1}) - \Phi(\mathbf{g}^*)\right] + c\varepsilon_t \\ &\leq \left(\frac{mL^2}{mL^2 + \mu^2}\right) \left[\left(\frac{mL^2}{mL^2 + \mu^2}\right) (\Phi(\mathbf{g}^{t-2}) - \Phi(\mathbf{g}^*)) + c\varepsilon_{t-1}\right] + c\varepsilon_t \\ &\cdots \\ &\leq \left(\frac{mL^2}{mL^2 + \mu^2}\right)^t (\Phi(\mathbf{g}^0) - \Phi(\mathbf{g}^*)) + c \sum_{s=1}^{t} \left(\frac{mL^2}{mL^2 + \mu^2}\right)^{t-s} \varepsilon_s \end{aligned}$$

□